

# Forward KL Can Be Better: Normalizing Flow Boltzmann Sampling with Log-Ratio Variation

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## Abstract

Forward KL training of a normalizing flow Boltzmann generator is mass-covering and usually trains well. Its weakness lies in the annealing chain that supplies its target samples, which starts at the flow’s own pushforward and mixes only among the modes the flow already covers. The objective can then miss the remaining modes while still reporting a high effective sample size, a fake ESS where the importance weights look benign but target mass is absent. The chain’s samples are also biased, and the forward KL controls only the mean of the log-ratio, so this bias passes into the trained flow. We address these failures with the log-ratio variation, a family of  $X$  functionals that penalize pairwise variation of the flow’s own log-ratio. Three weights give complementary roles: accuracy under the target, mode discovery against a wide-coverage measure built by quench and temper, and suppression of leakage outside the target. The balanced loss  $KL + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$  combines them at the cost of one extra sampling batch. A Fisher–Rao analysis shows that the forward KL still converges under the  $X_\mu$  term and that this term can lower the accuracy floor left by biased samples. The completed model problems confirm the intended accuracy and mode-coverage effects. On schedule-matched clock ladders, the balanced loss raises validation ESS on 36 of 37 shared levels and reduces the cumulative error-propagation factor by 2.7–4.2 across batch sizes.

## 1 Introduction

Drawing samples from a target distribution  $\mu \propto \exp(-U)$  on  $\mathbb{R}^d$ , known only up to its normalization constant, is a recurring problem in molecular simulation, Bayesian inference, and statistical physics. When  $\mu$  is high-dimensional and multimodal, its mass splits among basins of  $U$  separated by barriers. A local Markov chain Monte Carlo (MCMC) sampler that proposes small moves is then trapped: the time to cross a barrier of free-energy height  $\Delta F$  grows like  $e^{\Delta F}$ , so within any practical budget the chain explores the basin it began in and leaves the rest of the mass unseen (Metropolis et al., 1953; Hastings, 1970). The estimates such a chain returns are biased toward that basin. The bias is silent, since the chain reports nothing about the mass it never visited.

The classical responses soften the barriers rather than remove them. Parallel tempering (PT) and replica exchange run replicas at a ladder of temperatures and swap configurations between neighbors, carrying a basin crossed hot down to the temperature of interest (Swendsen and Wang, 1986; Geyer, 1991; Hukushima and Nemoto, 1996). Annealed importance sampling (AIS) bridges a tractable proposal to  $\mu$  along intermediate distributions while accumulating importance weights (Neal, 2001). Sequential Monte Carlo (SMC) carries a weighted particle population along the same kind of path (Doucet et al., 2001; Del Moral et al., 2006). Nested sampling recasts the normalizing integral as a one-dimensional quadrature over a rising likelihood threshold (Skilling, 2006). A line meeting the learned-sampler program partway replaces the fixed hot reference by a map, sharpening the swap schedule, tuning the endpoint, or training a flow to carry the hot reference toward the cold target (Syed et al., 2022; Surjanovic et al., 2022; Zhang et al., 2025). This line is the closest to ours; we differ in that our mode discovery is built off the ladder, rather than carried by the tempering chain itself. These methods are the workhorses of multimodal sampling and remain the references

against which a learned sampler is judged. They share one limit: they move mass only between basins a chain can already traverse within budget, so a basin no replica reaches contributes nothing, and raising the temperature enough to bridge every barrier flattens the structure one is trying to resolve (Woodard et al., 2009; Chehab et al., 2025). We do not discard this machinery; we use it. Our training samples come from an AIS ladder. The wide-coverage measure that drives mode discovery is built by a quench and temper construction that borrows the temperature-annealing idea while avoiding the reliance on a chain crossing every barrier; we supply the modes the chain cannot reach from a separate source.

Against this classical backdrop, Boltzmann generators (BG) replace the chain by a learned transport map (Noé et al., 2019). A normalizing flow is an invertible parametric map  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018). It is trained so that the pushforward  $\nu := (G^{-1})_{\#}\mu_0$  of an easily samplable reference  $\mu_0 \propto \exp(-U_0)$ , typically a Gaussian, approximates  $\mu$ . By convention  $G$  maps target to source,  $G_{\#}\mu \approx \mu_0$ , so  $G^{-1}$  draws a batch of independent samples by pushing  $\mu_0$  through it in one forward pass. Invertibility gives  $\nu$  in closed form by the change of variables, so each sample carries its own importance weight  $\mu/\nu$  against the unnormalized target. Reweighting then turns the samples into Monte Carlo estimates with controlled finite-sample error. The effective sample size (ESS) of those weights is a standard diagnostic of how well the flow matches  $\mu$  on the support it has reached (Liu, 2001; Doucet et al., 2001; Elvira et al., 2022).

The choice of training objective then divides the field. The original BG (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse Kullback–Leibler (KL) divergence

$$\text{(reverse)} \quad \text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx, \quad (1)$$

which needs only samples from  $\nu$  and energy evaluations, and is therefore cheap. It is also mode-seeking (Wainwright and Jordan, 2008): on multimodal  $\mu$  the flow tends to collapse onto a subset of the modes and cannot recover from a mode it never visits, since a missed mode contributes no gradient (Wu et al., 2020; Felardos et al., 2023). Several lines soften this without removing it, interleaving MCMC kernels with the deterministic layers, folding the symmetries of  $\mu$  into the architecture, lowering the gradient variance near the optimum, or annealing from a hot landscape toward  $\mu$  (Wu et al., 2020; Köhler et al., 2020; Vaitl et al., 2022; Schopmans and Friederich, 2025). None restores a dropped mode, because the reverse KL places no sample where  $\nu$  has no mass. Our  $X_{\bar{\nu}}$  term below plays the leakage-suppression role one wants from a reverse KL-like penalty, restraining the mass  $\nu$  places where  $\mu$  has none. It reads this from a stop-gradient copy of  $\nu$ , so the flow’s own sampling is never differentiated through, the source of the reverse KL’s instability on sharp targets.

The opposing objective minimizes the forward KL,

$$\text{(forward)} \quad \text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (2)$$

which is mass-covering, diverging wherever  $\nu$  misses mass that  $\mu$  carries (Wainwright and Jordan, 2008), but needs samples from  $\mu$ , the very object one is trying to produce. Existing work breaks this circularity by choosing where the target samples come from, and each choice fixes what the method can see: maximum likelihood on configurations from a reference thermodynamic state (Wirnsberger et al., 2020), a flow trained on a running chain and used as a global proposal (Gabrié et al., 2022), fitting to a molecular dynamics (MD) trajectory (Klein et al., 2023), amortizing coverage across many molecules through a shared architecture (Klein and Noé, 2024), or supplying the intractable gradient from a persistent chain or an  $\ell_2$  fit to coupled targets (Naesseth et al., 2020; Rehman et al., 2026). In every case the forward KL is silent about mass its sample source never reaches: a mode absent from the reference data, the running chain, or the trajectory set leaves no trace in the gradient, as a missed mode is invisible to a chain. These remedies change the training signal or amortize coverage across systems. By contrast we keep the bare forward KL and confront the circularity directly, adding the log-ratio variation below, which reuses the same samples on a single target. Our AIS ladder supplies the target samples. Because that ladder inherits the classical limit, the regularizer draws on an independent sample source built to cover modes the chain misses.

The line closest to ours couples an annealing ladder to flow training so the flow improves the sampler that trains it. The flow annealed importance sampling bootstrap (FAB) (Midgley et al.) replaces the forward KL

by the  $\alpha$ -divergence at  $\alpha = 2$ , whose integrand is proportional to  $\mu^2/\nu$  and whose minimizer also minimizes the variance of the importance weights. It draws training points from AIS chains targeting this minimum-variance surrogate and recycles them through a prioritized replay buffer. Related work weaves a learned flow into each stage of an SMC annealing path, reweights down a temperature ladder, adds an inference-time SMC correction at molecular scale, or curbs the mass teleportation these geometric paths allow (Arbel et al., 2021; Matthews et al., 2022; Schopmans and Friederich, 2025; Tan et al., 2025; von Klitzing et al., 2026). We take this AIS-coupled forward KL line as our point of departure and depart from it on three counts, set side by side in Remark 3. First, we keep the bare forward KL and add the log-ratio variation rather than change the divergence, so the minimizer is unchanged and the regularizer reuses the same samples and the same autograd graph. Second, our AIS ladder targets  $\mu$  itself and is score-free in the flow: every chain moves under the target score  $-\nabla U$ , the flow enters only through evaluations of the log-ratio, and no replay buffer is kept. By contrast, the surrogate  $\mu^2/\nu$  has a score that carries the positional gradient of the pushforward density at every chain step. Third, mode discovery is delegated to a quench and temper measure  $\hat{\mu}$  rather than asked of a chain that would still have to cross the barriers of  $\mu$ . The trade-offs are real. The order-2 Rényi divergence behind  $\mu^2/\nu$  carries a stronger tail-coverage incentive than the forward KL, and with a sufficient annealing budget FAB needs no separate wide-coverage construction. Where an inference-time SMC pass buys scale, we keep the correction inside the tractable per-sample importance weight, so the score-free ladder and the absent replay buffer leave a small memory footprint and let the method run on a single consumer GPU. The off-policy log-dispersion regularizer penalizes a scalar, variance-type spread of  $\log(\mu/\nu)$  (Schopmans et al., 2026). The log-ratio variation below instead penalizes the localized pairwise variation of  $\log(\mu/\nu)$  under a chosen weight. This weight lets one family take three complementary roles from a single pairwise integral: mode discovery through  $\hat{\mu}$ , leakage suppression through  $\bar{\nu}$ , and shape control through  $\mu$ .

A separate tradition drops the bijection and trains a stochastic process directly: a controlled stochastic differential equation, a denoising diffusion, or a flow-matching velocity carries the reference to  $\mu$  along a continuous-time path, learning the drift rather than an invertible map (Zhang and Chen, 2022; Vargas et al., 2023; Berner et al., 2024; Nüsken and Richter, 2021; Richter and Berner, 2024). Energy-only variants such as iterated denoising energy matching (iDEM) alternate a sampling loop with a score- or velocity-fitting loop and a replay buffer (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024). The non-equilibrium transport sampler augments AIS with a learned drift and so sits between the AIS-coupled and diffusion lines while still offering no tractable terminal density (Albergo and Vanden-Eijnden, 2025), and successors scale the idea to larger many-body systems (Havens et al., 2025; Akhound-Sadegh et al., 2025). These samplers are expressive and train from the energy alone. The cost is decisive for our purpose: once the bijection is gone, the terminal density  $\nu$  is no longer available in closed form, so the importance weight  $\mu/\nu$ , the ESS read from it, and the reweighting that corrects residual bias are out of reach without a further per-sample ordinary-differential-equation solve (He et al., 2025). We commit to invertible flows for that reason: the invertibility of  $G$  keeps the log-ratio  $\log(\mu/\nu)$  exact and inexpensive per sample, and with it the finite-sample importance-sampling pipeline on which both the forward KL and the log-ratio variation rest. The two lines are complementary rather than competing: one buys scale through dynamics at the cost of a tractable density, the other keeps the density and the reweighting at the cost of invertibility. Table 1 places these training lines side by side.

|            | objective                                     | AIS<br>required | importance<br>weights | mode<br>source         | replay<br>buffer | score-free<br>flow |
|------------|-----------------------------------------------|-----------------|-----------------------|------------------------|------------------|--------------------|
| reverse KL | $\text{KL}(\nu\ \mu)$                         | —               | ✓                     | none                   | —                | ✓                  |
| forward KL | $\text{KL}(\mu\ \nu)$                         | ✓               | ✓                     | AIS chain              | —                | ✓                  |
| FAB        | $\alpha=2$ divergence, $\mu^2/\nu$            | ✓               | ✓                     | AIS chain              | ✓                | —                  |
| iDEM       | energy matching                               | —               | —                     | sampling loop          | ✓                | —                  |
| ours       | $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$ | ✓               | ✓                     | $\hat{\mu}$ , no chain | —                | ✓                  |

Table 1: Where the balanced  $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$  flow sits among Boltzmann generator training lines. Entries are qualitative; ✓marks a property the line has by construction, — its absence.

**What is new.** The forward KL inherits the limits of its target surrogate: an annealing procedure initialized at  $\nu$  may mix only among modes  $\nu$  already covers. This leaves two core weaknesses, together with a third mechanism the same regularizer addresses. The first is silent missed modes, the fake ESS pitfall. If  $\nu$  misses a mode of  $\mu$ , the sampler initialized at  $\nu$  may remain in the reached basins; its observed importance weights can be nearly uniform and the ESS close to one while target mass is absent (Noé et al., 2019; Midgley et al.). The second is accuracy under an imperfect target surrogate. Finite batches add noise, and the score-free annealing procedure of Section 3.2 is biased by construction. Because the forward KL constrains only the mean of  $\log(\mu/\nu)$  under that surrogate and leaves its spread free, the approximation error passes directly into the trained flow. The third mechanism is leakage: where  $\mu$  vanishes, the forward KL integrand  $\mu \log(\mu/\nu)$  vanishes too, so the gradient carries no signal about mass that  $\nu$  places outside the support of  $\mu$ . Better calibration on the modes already reached does not cure a missed mode.

Our remedy is to regularize the forward KL with the *log-ratio variation*, a family of X functionals  $X_\omega$  named for the pairwise variation of the log-ratio  $z = \log(\mu/\nu)$  they penalize. For a non-negative weight  $\omega$  on  $\mathbb{R}^d$ , not necessarily normalized, let

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x) \omega(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (3)$$

against which both arguments are integrated. The idea is direct: matching  $\nu$  to  $\mu$  is the same as flattening  $z$ , and the pairwise difference cancels the unknown normalizer of  $\mu$ . Where the forward KL constrains  $\nu \approx \mu$  only in a global, Jensen-averaged sense,  $X_\omega$  penalizes the variation of  $z$  wherever  $\omega$  places mass and localizes the constraint to the support of  $\omega$ . It vanishes whenever  $z$  is constant on that support. In particular, adding this non-negative term to the exact forward KL preserves  $\nu = \mu$  as a global minimizer. For  $\omega = \mu$  its integrand is built from the same log-ratio field as the forward KL, reusing the same samples and the same autograd graph.

Three choices of  $\omega$ , with complementary roles, jointly address these failures. With  $\omega = \mu$ , the functional  $X_\mu$  controls the spread of  $z$  under the target where the forward KL controls only its mean, tightening the density match once  $\nu$  has reached the modes and trimming the bias inherited from imperfect annealed samples. With  $\omega = \hat{\mu}$ , a coarse, over-spread approximation of  $\mu$ , the functional  $X_{\hat{\mu}}$  drives mode discovery: by the insensitivity above, the only role of  $\hat{\mu}$  is to place mass on modes and tails the current  $\nu$  has not yet covered, and the gradient then pulls the flow toward them. We build  $\hat{\mu}$  once, before training, with a quench and temper construction that bridges basins, collapses onto mode centers, and rejuvenates with a short MALA chain. It errs toward broad coverage, and inherits a blind spot on heavy outer tails. With  $\omega = \bar{\nu}$ , a stop-gradient copy of the current pushforward, the functional  $X_{\bar{\nu}}$  plays the reverse KL role without its instability, penalizing the mass  $\nu$  places outside the support of  $\mu$  and so suppressing the leakage the other two terms cannot see. Our preferred loss combines all three with the forward KL,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) + X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu), \quad (4)$$

with  $\lambda \geq 0$  and either  $\alpha = \beta = 0$  or  $\alpha, \beta \geq 0$  with  $\alpha + \beta = 1$ . The third term is  $X_\omega$  at the mixture weight  $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$ . Putting the mixture inside a single pairwise integral does more than add the two self-terms: expanding the product measure also generates a cross term that anchors the in-support and out-of-support log-ratios to each other. Written through the change of variables, every term reads off one scalar field along the flow,

$$z(y) = U_0(G(y)) - U(y) - \log |\det J_G(y)|, \quad (5)$$

up to an additive constant that cancels in each pairwise difference, so the loss in terms of the flow map  $G$  is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu} [|z(y) - z(y')|]}_{X_\mu(\mu\|\nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu} + \beta\bar{\nu}} [|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}. \quad (6)$$

The  $X_\mu$  term reuses the target batch and log-ratio evaluations of the forward KL, while the mixture term draws and evaluates one further batch. Adding either term preserves  $\nu = \mu$  as a global minimizer of the exact forward KL.

**Contributions.** This paper makes the following contributions. We introduce the log-ratio variation  $X_\omega$ , a family of  $X$  functionals that augment the forward KL by penalizing the pairwise variation of the flow’s own log-ratio  $z = \log(\mu/\nu)$ . We study three instances in complementary roles,  $X_\mu$ ,  $X_{\hat{\mu}}$ , and  $X_{\bar{\nu}}$ , together with the balanced loss  $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ , which folds accuracy, mode discovery, and leakage suppression into one objective. The construction keeps the AIS target score free of the flow and requires no replay buffer; the balanced mixture adds one sampling batch beyond a plain forward KL step. We give a Fisher–Rao analysis showing that the forward KL decays along the  $\text{KL} + \lambda X_\mu$  flow under no structural assumption on  $\mu$  and, under a bound on  $\lambda$ , that the  $X_\mu$  term lowers the accuracy floor left by biased target samples. We pair the ESS with a coverage diagnostic (Naeem et al., 2020), so density calibration on the reached support is not confused with mode coverage. Finally, we build an adaptive-temperature Boltzmann generator that grows its temperature ladder from per-stage sampling diagnostics. The completed numerical study isolates the three roles of the  $X$  functionals on smooth and collective-barrier targets and verifies the balanced loss on schedule-matched clock ladders; only the molecular production results remain explicitly pending.

**Organization.** Section 2 states the main Fisher–Rao gradient-flow results for the regularized loss, with the full derivations, convergence proof, and biased-target accuracy analysis deferred to Appendix A. Section 3 turns the loss into a training algorithm for the flow map  $G$ , detailing the annealing procedure that approximates  $\mu$ , the quench and temper construction of the wide-coverage measure  $\hat{\mu}$ , and the ESS and coverage diagnostics. Section 4 builds the adaptive-temperature Boltzmann generator on top of this loss. Section 5 reports the numerical study on model problems, and Section 7 concludes.

## 2 Fisher–Rao gradient flow analysis

Section 1 introduced the forward KL with log-ratio variation. Before turning it into a training algorithm in Section 3, we summarize its gradient-flow properties on the space of densities; Appendix A gives the full derivations and proofs. This continuous-time population idealization exposes the mechanism of the loss but does not imply a convergence rate for the discrete optimizer: we view  $\nu$  as a probability density evolving along the gradient flow and ask how fast  $\text{KL}(\mu\|\nu_t)$  decays to zero.

For a tractable analysis we keep only the forward KL and the shape term  $X_\mu$ , the log-ratio variation at weight  $\omega = \mu$ . The mode-discovery and leakage terms  $X_{\hat{\mu}}$  and  $X_{\bar{\nu}}$  are left out: their wide-coverage and stop-gradient weights are training devices that do not enter this density-space idealization. The loss is then

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu), \quad (7)$$

with  $\lambda > 0$  fixed. Write  $w = \mu/\nu$  for the importance weight and  $z = \log w$  for the log-ratio. The statements below apply along any positive solution of the Fisher–Rao flow; for  $\lambda \leq \frac{1}{2}$ , the explicit birth term in Appendix A is non-negative and preserves positivity.

The choice of Riemannian metric matters. In the Wasserstein-2 geometry the forward KL is not displacement convex, so the Otto–Villani / Bakry–Émery machinery that yields exponential convergence for the reverse KL does not apply. We therefore work in the Fisher–Rao (FR) geometry, in which the forward KL admits an unconditional exponential decay and  $X_\mu$  adds a nonlocal birth–death modulation: the update at a point is driven by how its importance weight ranks against the population of weights under  $\mu$ . Writing the  $\nu$ -mean as  $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$ , the FR gradient of a functional  $\mathcal{F}$  and the flow it generates are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left( \frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{L}(\nu_t), \quad (8)$$

and the mean-subtraction makes  $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$ , so the flow conserves probability mass.

Differentiating  $\text{KL}(\mu\|\nu_t)$  along this flow, and reducing the  $X_\mu$  term by a rank-correlation identity, gives the dissipation

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = -\chi^2(\mu\|\nu_t) - \lambda \mathbb{E}_{x, y \sim \mu} \left[ \left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right], \quad (9)$$

with  $\chi^2$  the chi-squared divergence. The regularizer therefore contributes an additional non-positive term to the instantaneous KL dissipation. Since  $\chi^2(\mu\|\nu_t) \geq \text{KL}(\mu\|\nu_t)$ , Grönwall’s lemma yields the exponential



rate

$$\text{KL}(\mu\|\nu_t) \leq e^{-t} \text{KL}(\mu\|\nu_0), \quad t \geq 0, \quad (10)$$

under no structural assumption on  $\mu$  — no log-concavity, spectral gap, or growth condition. The rate does not claim forward KL is intrinsically faster. It is a population statement on the space of densities, agnostic to the cost of producing the target samples that AIS must supply. The contribution of  $X_\mu$  to (9) is structural — it amplifies mass where  $\nu_t$  undercovers  $\mu$  — rather than a faster transient.

The convergence above assumes the outer expectation is taken under the exact target. In practice AIS supplies only an approximation  $\tilde{\mu}$ , and the realized loss replaces  $\mu$  by  $\tilde{\mu}$  in its outer integrands. At  $\lambda = 0$  a stationary point is  $\nu^* = \tilde{\mu}$ : pure forward KL inherits the AIS bias verbatim, leaving the residual error  $\text{KL}(\mu\|\tilde{\mu})$  as a hard floor. The shape term lowers this floor.

**Theorem 1** *Assume  $\mu$  is absolutely continuous with respect to  $\tilde{\mu}$  and  $\lambda \in [0, \frac{1}{2})$ . Then every stationary density  $\nu^*$  of the biased-surrogate loss  $\tilde{\mathcal{L}}$  defined in (36) satisfies*

$$\text{KL}(\mu\|\nu^*) \leq \text{KL}(\mu\|\tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (11)$$

The rank-based feedback inflates  $\nu^*$  where it undercovers  $\mu$  and deflates it elsewhere, pulling the stationary density off  $\tilde{\mu}$  back toward  $\mu$  by an amount set by the spread of the importance weights  $\mu/\nu^*$  across the support of  $\tilde{\mu}$ . Thus  $X_\mu$  is not merely a stabilizer for the forward KL flow: within the stated range of  $\lambda$ , it lowers the asymptotic accuracy floor left by biased target samples, a correction especially relevant at small batches and early stages. Our experiments use  $\lambda = 1$  and show robust performance there, but the theorem itself guarantees the bound only for  $\lambda < \frac{1}{2}$ . The full FR-gradient derivation, the rank-correlation lemma, the convergence proof, and the accuracy proof are in Appendix A. This analysis lives on the space of densities and assumes access to  $\mu$ ; the next section supplies approximations of  $\mu$  and the wide-coverage measure  $\hat{\mu}$  that make the loss trainable.

### 3 Flow training with log-ratio variations

This section turns the abstract loss of the introduction into a training algorithm for the flow map  $G$ . Let  $U_0$  and  $U$  be the source and target potentials on  $\mathbb{R}^d$ , so that  $\mu_0 \propto \exp(-U_0)$  is the samplable reference and  $\mu \propto \exp(-U)$  is the target. We parameterize  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  so that  $G_{\#}\mu \approx \mu_0$  and write  $\nu := (G^{-1})_{\#}\mu_0$  for the induced approximation to  $\mu$ . The loss itself evaluates only the native target-to-source map  $G$  and its log-Jacobian; it does not explicitly involve  $G^{-1}$ . The inverse map is nevertheless used to initialize AIS and to draw model samples. These calls are inference-only evaluations outside the autodiff graph and remain reasonably inexpensive. We first assemble the training loss and identify its sampling ingredients (Section 3.1), then supply them in turn: a score-free annealing approximation of  $\mu$  (Section 3.2), the quench and temper construction of the wide-coverage measure  $\hat{\mu}$  (Section 3.3), and the ESS and coverage diagnostics used to evaluate the trained flow (Section 3.4).

#### 3.1 The log-ratio-variation training loss

By the change-of-variables formula the pullback density of  $\mu_0$  under  $G$  is  $\nu(y) = \mu_0(G(y)) |\det J_G(y)|$ , so the log-ratio between the target and the model along the flow is the single scalar field  $z(y)$  of (5). Both ingredients of the loss are functionals of  $z$ . The forward KL is its mean under the target,  $\text{KL}(\mu\|\nu) = \mathbb{E}_{y \sim \mu}[z(y)] + \text{const}$  (cf. (2)). Each log-ratio variation  $X_\omega$  (3) is the mean absolute pairwise variation of  $z$  under a non-negative weight  $\omega$ ,  $X_\omega(\mu\|\nu) = \mathbb{E}_{y, y' \sim \omega}[|z(y) - z(y')|]$  when  $\omega$  is normalized (in general it is the double integral (3)). Combining the shape term  $\omega = \mu$  with the balanced mixture  $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$  that handles mode discovery and leakage suppression, the total training loss in terms of the flow map  $G$  is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu}[z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu}[|z(y) - z(y')|]}_{X_\mu(\mu\|\nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu} + \beta\bar{\nu}}[|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}. \quad (12)$$

We refer to (12), previewed as (6), as forward KL with log-ratio variation. The balanced method used in the staged experiments is named by its loss,  $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ . Note the two pushforward measures in

(12). The model distribution  $\nu = (G^{-1})_{\#}\mu_0$ , with density  $\nu(y) = \mu_0(G(y))|\det J_G(y)|$ , is the flow’s live approximation to the target  $\mu$  and depends on the trainable parameters of  $G$ . The barred measure  $\bar{\nu}$  is the *same* pushforward frozen at the current parameters, a stop-gradient copy of  $\nu$ , which enters the mixture term only as a sampling weight and is never differentiated through.

The three sampling measures entering (12) address complementary failure modes of the bare forward KL:

- $\mu$ : *shape accuracy and robustness*. The shape term  $X_\mu$  compares log-importance ratios pairwise across the target batch and lowers the asymptotic accuracy floor under biased target samples (Section 2). Small or noisy batches are its primary design regime; difficult high-dimensional or staged targets can retain an aggregate advantage at larger batches.
- $\hat{\mu}$ : *mode discovery*. The wide-coverage measure built by quench and temper (Section 3.3) places sample pairs across modes that the score-free annealing surrogate did not reach within its budget, pulling  $\nu$  toward mass the bare loss never sees.
- $\bar{\nu}$ : *leakage suppression*. The frozen pushforward penalizes the mass the flow places outside the support of  $\mu$ , preventing the overfit that target samples alone cannot detect: where  $\mu$  provides no samples, the bare forward KL provides no signal.

Adding these non-negative terms to the exact forward KL preserves  $\nu = \mu$  as a global minimizer. Their purpose is to improve numerical robustness without changing the desired population solution.

**Remark (Robustness across regimes)** *As discussed in Section 2, the exponential rate (10) is a statement about the idealized FR flow. The discrete Adam dynamics carries no corresponding speed guarantee. The loss was designed primarily to stabilize training with small or imperfect target batches. Whether its benefit persists through every level of a difficult staged target is a separate empirical question. The clock experiment therefore uses an exact schedule-matched comparison and is reported only after every paired level and downstream occupancy diagnostic has been regenerated.*

The forward KL and  $X_\mu$  share one target batch: evaluate  $\mathbf{z} = (z(y_1), \dots, z(y_B))$ , take  $\text{mean}(\mathbf{z})$  for the forward KL, and  $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$  for a random permutation  $\sigma$ . Thus  $X_\mu$  adds only a permutation and pairwise difference to the same autograd graph. The balanced mixture evaluates one additional batch drawn from  $(\hat{\mu} + \bar{\nu})/2$  and applies the same randomized pairwise estimator.

The loss draws on three measures, of which only two must be built. The  $\bar{\nu}$  batch is the raw pushforward  $\bar{y}_i = G^{-1}(x_i)$  of source samples  $x_i \sim \mu_0$ , taken with the gradient through  $G$  detached. This avoids differentiating through the sampled pushforward. The other two require construction: an approximation of the target  $\mu$  for the forward KL and  $X_\mu$  terms, supplied by the annealing procedure of Section 3.2, and samples of the wide-coverage measure  $\hat{\mu}$  for the mixture term, supplied by quench and temper (Section 3.3). With equal target and mixture batch sizes,  $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$  uses about twice the raw per-step sample evaluation of forward KL. Inference uses the same flow architecture for both methods.

We use two fixed settings. The  *$X_\mu$ -only setting*  $\lambda = 1, \alpha = \beta = 0$  adds only  $X_\mu$  to the forward KL and reuses its target batch and autograd graph. The *balanced setting*  $\lambda = 1, \alpha = \beta = \frac{1}{2}$  draws the mixture batch equally from  $\hat{\mu}$  and  $\bar{\nu}$ : the  $\hat{\mu}$  half pulls  $\nu$  toward modes discovered by quench and temper that the annealing procedure did not reach within its budget, while the  $\bar{\nu}$  half restrains mass leaked between or outside target modes. We use this balanced setting whenever the source  $\mu_0$  may not cover all modes of  $\mu$  and report it as  $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ .

### 3.2 Approximating $\mu$ by score-free annealing

Evaluating the  $\mu$ -expectations in (12) requires samples from  $\mu$ , which are not directly available. The natural surrogate draws  $y_1, \dots, y_B$  from the pushforward  $\nu = (G^{-1})_{\#}\mu_0$  and reweights them by

$$w(y) := \exp(z(y)) = \exp\left(U_0(G(y)) - U(y) - \log|\det J_G(y)|\right), \quad y \in \mathbb{R}^d, \quad (13)$$

followed by a resampling step and a short Metropolis-adjusted Langevin algorithm (MALA) simulation to diffuse the duplicates. In high dimensions, however,  $\mu$  and  $\nu$  can differ greatly, so the weights  $w(y_i)$  collapse

onto a few outliers and the samples are of low quality. We temper the reweighting along the geometric path of  $M + 1$  distributions

$$\pi_m(y) \propto \nu(y)^{1-\frac{m}{M}} \mu(y)^{\frac{m}{M}} \propto ((G^{-1})_{\#} \mu_0)(y) w(y)^{\frac{m}{M}}, \quad m = 0, 1, \dots, M, \quad (14)$$

so that  $\pi_0 = \nu$  and  $\pi_M = \mu$ . Each step reweights by only  $w(y)^{\frac{1}{M}}$ , reducing the degeneracy of direct importance sampling.

---

**Algorithm 1:** Score-free annealing from  $(G^{-1})_{\#} \mu_0$  toward  $\mu$

---

**Input:** samples  $y_1, \dots, y_B$  from  $(G^{-1})_{\#} \mu_0$ , ladder length  $M$ , intermediate distributions  $\pi_m$  (14)

**Output:** a biased sample approximation  $y_1, \dots, y_B$  of  $\mu$

```

1 for  $m = 1, \dots, M$  do
    // reweight along the geometric path
2   Compute the incremental weights on  $y_1, \dots, y_B$ ,  $w(y_i)^{\frac{1}{M}} = \exp(z(y_i)/M)$ ;
    // resample
3   Resample from the weighted set,  $y'_1, \dots, y'_B \leftarrow \text{resample}(y_{1:B}, (w(y_i)^{\frac{1}{M}})_{i=1}^B)$ ;
    // rejuvenate under the target
4   Diffuse the resampled duplicates by a short MALA chain on  $\mu$ ,  $y_{1:B} \leftarrow \text{MALA}(y'_{1:B}, U)$ ;
5 end
```

---

An exact SMC sampler on the path (14) would rejuvenate at step  $m$  with a Markov kernel invariant for  $\pi_m$ . Its score contains  $\nabla_y \log \nu(y)$  and hence positional derivatives of the flow density, including  $\nabla_y \log |\det J_G(y)|$ . To keep the training sampler score-free in the flow, Algorithm 1 instead applies MALA directly under  $\mu$ , using only  $-\nabla U$  in its proposal. MALA preserves  $\mu$ , but it does not preserve the intermediate  $\pi_m$ , so the output is a biased target surrogate rather than an exact AIS/SMC sample. The construction trades exact intermediate invariance for a cheap sampler that pulls resampled particles toward the final target. Section 2 models precisely this situation through the surrogate measure  $\tilde{\mu}$ , and the numerical study evaluates the resulting trained flow by independent importance weights and coverage diagnostics.

**Remark (Choosing the ladder length  $M$ )** For simple targets close to the source,  $M = 1$  suffices. The choice  $M = d$  is typical but not necessary in every test — an empirical rule by which the ladder length scales with the dimension of the deformation the flow must bridge, of which  $d$  is the worst case. For instance, annealing  $\mu_0 \propto e^{-|x|^2}$  to  $\mu \propto e^{-2|x|^2}$  on  $\mathbb{R}^d$  deforms all  $d$  coordinates at once, and a ladder of  $M = d$  steps keeps the weighted empirical distribution at a finite ESS that does not degrade with  $d$ . When the deformation concentrates on fewer directions, a much shorter ladder suffices.

**Remark (Comparison with FAB)** FAB (Midgley et al.) also couples AIS to flow training, but places the flow inside the chains. Its transitions target the mode-seeking surrogate  $\mu^2/\nu$ , whose score

$$\nabla_y \log \frac{\mu^2(y)}{\nu(y)} = -2\nabla U(y) - \nabla_y \log \nu(y)$$

carries the positional gradient of the pushforward density at every MCMC step, the term avoided by MALA under  $\mu$  above. Training then feeds on a replay buffer of past AIS samples reweighted against staleness. The MALA transition score in our scheme is flow-free: every MALA chain targets  $\mu$  with score  $-\nabla U$ . Within the annealing sampler, the flow enters through the initial  $G^{-1}$  proposal and evaluations of the log-ratio  $z$ ; each gradient step runs Algorithm 1 afresh on a new source batch, so no buffer is kept. The one persistent object, the  $N$ -sample quench and temper set of Section 3.3, is built once and covered by sample robustness. FAB explores through its AIS chain, which must still cross the barriers of  $\mu$ ; here mode discovery is delegated to the melt and quench before MALA tempering. The trade-offs are real: the order-2 Rényi divergence behind  $\mu^2/\nu$  carries a stronger tail-coverage incentive than the forward KL, and with sufficient AIS budget FAB needs no wide-coverage construction.



### 3.3 Constructing $\hat{\mu}$ by quench and temper

The mixture term needs samples from the wide-coverage measure  $\hat{\mu}$ . Let  $\mathcal{Y} = \{x_1, \dots, x_N\}$  be the complete validation population, drawn from the source measure for the flow being trained. For a single-flow problem QT is applied to  $\mathcal{Y}$  once before training; in the staged generator of Section 4, it is applied to the complete population  $\mathcal{Y}_{k-1}$  once per stage attempt. The resulting set is held fixed during optimization. At each training step a mini-batch of  $B$  samples is drawn from it and refreshed by a short MALA chain before entering the  $X_\omega$  computation of Section 3.1 with  $\mu$  replaced by  $\hat{\mu}$ . Quench and temper (QT) is the map

$$\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}, U)$$

from the complete  $N$ -particle validation population and the target potential  $U$  to a wide-coverage set  $\hat{\mathcal{Y}} = \{\hat{x}_1, \dots, \hat{x}_N\}$  distributed as  $\hat{\mu}$ . It proceeds in three phases: diffusion (melt), optimization (quench), and MALA rejuvenation (temper).

---

**Algorithm 2:** Quench and temper (QT): proactive mode discovery

---

**Input:** complete validation population  $\mathcal{Y} = \{x_1, \dots, x_N\}$ , target potential  $U$ , melt scale  $m_e$   
**Output:** wide-coverage set  $\hat{\mathcal{Y}} = \{\hat{x}_1, \dots, \hat{x}_N\}$  forming  $\hat{\mu}$ , written  $\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}, U)$   
 // diffusion (melt)  
 1 Scatter the samples to a wider region of  $\mathbb{R}^d$ ,  $x_i \leftarrow x_i + m_e \zeta_i$  with  $\zeta_i \sim \mathcal{N}(0, I_d)$  i.i.d. and melt scale  $m_e > 0$ ;  
 // optimization (quench)  
 2 Drive each sample toward a mode center of  $\mu$  by L-BFGS on  $U$ , updating  $x_1, \dots, x_N$ ;  
 // rejuvenation (temper)  
 3 Spread the samples around each mode,  $\hat{x}_{1:N} \leftarrow \text{MALA}(x_{1:N}, U)$ ;

---

The output samples concentrate near mode centers of  $\mu$ , while the initial Gaussian scatter broadens the set of basins the quench can reach. The  $X_{\hat{\mu}}$  term therefore supplies training pairs on modes where the current pushforward  $\nu$  places too little mass, including modes the score-free annealing procedure did not reach within its allotted budget.

A large melt scale  $m_e$  can bias the QT set toward modes far from the bulk of the input population. This is intentional:  $\hat{\mu}$  is a discovery measure rather than a calibrated approximation of target mode weights. Adding  $X_{\hat{\mu}}$  to the exact forward KL still preserves  $\nu = \mu$  as a global minimizer, while the empirical mode weights in the finite  $N$ -sample QT set can affect the optimization path.

### 3.4 Importance sampling at evaluation: ESS and coverage

At evaluation time the trained flow  $G$  supplies the model distribution  $\nu = (G^{-1})_{\#}\mu_0$  as an approximation to  $\mu$ , together with the tractable density  $\nu(y) = \mu_0(G(y))|\det J_G(y)|$  that any normalizing flow makes available. The importance weight  $w(y) = \exp(z(y))$  of (13) can therefore be evaluated on any pushforward sample without fitting an additional density model. Drawing  $\mathbf{y} = (y_1, \dots, y_N)$  with  $y_i = G^{-1}(x_i)$  and  $x_i \sim \mu_0$  i.i.d., the weights  $w(y_i)$  serve two uses. First, they diagnose how well  $\nu$  approximates  $\mu$ , through the effective sample size defined below. Second, they correct residual bias by reweighting or resampling the pushforward, followed when desired by a short MALA refresh under  $\mu$ . The better the flow, the more uniform the weights and the less correction the resampling performs.

We assess this pipeline through two metrics: the effective sample size, which measures how uniform the importance weights are on the support reached by the flow, and the coverage, which checks whether every mode of  $\mu$  has been reached at all.

The first metric is the effective sample size, normalized to  $[0, 1]$  by dividing the classical ESS by the sample count  $N$ . Let  $\mathbf{y} = (y_1, \dots, y_N) \in \mathbb{R}^{N \times d}$  be i.i.d. samples from  $\nu$ , with importance weights  $w(y_i)$  given

by (13). The ESS is defined as

$$\text{ESS}(\mathbf{y}) = \frac{\left(\sum_{i=1}^N w(y_i)\right)^2}{N \sum_{i=1}^N w(y_i)^2} \in [0, 1], \quad (15)$$

where the upper bound 1 follows from the Cauchy–Schwarz inequality and is attained when all weights are equal. The factor  $N$  in the denominator distinguishes our normalized ESS from the classical effective sample size  $(\sum_i w(y_i))^2 / \sum_i w(y_i)^2 \in [1, N]$  standard in the literature (Elvira et al., 2022): that quantity counts an effective number of samples and scales with the batch, whereas dividing by  $N$  leaves the per-sample efficiency, a fraction in  $[0, 1]$  comparable across batch sizes. When  $G$  is perfectly trained the weights are constant and the ESS reaches 1. As noted in the introduction, however, this conclusion is one-way: a high ESS indicates that the sampled weights are uniform *on the support reached by  $\nu$* , but modes of  $\mu$  that the pushforward never reaches contribute nothing to the sum and so cannot lower the ESS, which gives rise to the *fake ESS* pitfall.

To detect such missing modes, we pair the ESS with the coverage metric of Naeem et al. (2020). Let  $\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_N) \in \mathbb{R}^{N \times d}$  be the samples of  $\hat{\mu}$  produced by the quench and temper algorithm (Algorithm 2). For each  $1 \leq i \leq N$ , the  $\kappa$ -nearest-neighbor radius of  $\hat{y}_i$  (with  $\kappa = 5$ ) is

$$\text{NND}_\kappa(\hat{y}_i; \hat{\mathbf{y}}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq N : 0 < |\hat{y}_i - \hat{y}_j| < r\} \geq \kappa \right\}, \quad (16)$$

i.e., the distance from  $\hat{y}_i$  to its  $\kappa$ -th nearest neighbor in  $\hat{\mathbf{y}}$ . The coverage of  $\hat{\mathbf{y}}$  by the pushforward samples  $\mathbf{y}$  is then

$$\text{Coverage}_\kappa(\mathbf{y}; \hat{\mathbf{y}}) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - \hat{y}_i| \leq \text{NND}_\kappa(\hat{y}_i; \hat{\mathbf{y}}) \right\}, \quad (17)$$

i.e., the fraction of points in  $\hat{\mathbf{y}}$  whose  $\kappa$ -nearest-neighbor ball contains at least one pushforward sample. A coverage close to 1 indicates that the finite QT reference set is well covered by the flow, while a coverage well below 1 exposes missing regions that the ESS alone cannot see. Because  $\hat{\mu}$  is the same wide-coverage measure that  $X_{\hat{\mu}}$  trains toward, this diagnostic checks only the modes found by QT; it cannot reveal a mode absent from the reference set.

## 4 Adaptive-temperature Boltzmann generator

Section 3 introduced forward KL with log-ratio variation (12). For a difficult high-dimensional or strongly multimodal target, training one flow map directly from the source  $\mu_0 \propto \exp(-U_0)$  to the target  $\mu \propto \exp(-U)$  can be impractical. We instead decompose the transport into a sequence of easier stages along a temperature ladder. Each flow learns one stage, a controlled change in the potential, and the ladder is built adaptively from sampling diagnostics.

### 4.1 Construction of temperature ladder

To bridge the simple source  $\mu_0 \propto \exp(-U_0)$  to the target  $\mu \propto \exp(-U)$ , we seek an increasing sequence

$$0 = t_0 < t_1 < \dots < t_K = 1$$

defining a sequence of intermediate distributions

$$\mu_k \propto \exp(-U_{t_k}), \quad U_{t_k}(x) = (1 - t_k) U_0(x) + t_k U_1(x), \quad k = 0, 1, \dots, K,$$

with  $U_1 := U$  an alias for the target potential, so that  $\mu_0$  is the source and  $\mu_K$  the target, together with  $K$  flow maps  $\{G_k\}_{k=1}^K$  trained so that  $(G_k^{-1})_\# \mu_{k-1} \approx \mu_k$ . The number of stages  $K$  is flexible. The design choice is each increment  $\Delta t_k = t_k - t_{k-1}$ . Too large an increment degrades the training of stage  $k$ ; too small an increment wastes computation. We want the largest step each flow can still learn well.

Training the flow  $G_k$  requires a target surrogate for  $\mu_k$ , supplied during optimization by Section 3.2. An optional classical SMC gate can screen the outer increment  $\Delta t_k$  before training. Its smallest per-step ESS is a heuristic measure of how difficult that candidate bridge is for the current particle population. The

---

**Algorithm 3:** Adaptive selection of the temperature step  $\Delta t_k$ 

---

**Input:** potentials  $U_0, U$ , previous level  $t_{k-1}$ , validation population  $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\}$  approximating  $\mu_{k-1}$ , SMC ESS floor  $\tau_{\text{SMC}} = 0$ , shrink factor  $\gamma = 0.7$ , initial guess  $t_{\text{init}}$

**Output:** next level  $t_k$ , giving the step  $\Delta t_k = t_k - t_{k-1}$

```
1  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U_1;$   
   // start from the supplied initial guess  
2  $t_k \leftarrow t_{\text{init}};$   
3 if  $\tau_{\text{SMC}} = 0$  then  
4   | return  $t_k;$   
5 end  
6 while true do  
   // run classical SMC across the candidate step  
7   Run  $M$ -step classical SMC on a working copy of all  $N$  particles in  $\mathcal{Y}_{k-1}$ , using the linear  
   potential bridge from  $U_{t_{k-1}}$  to  $U_{t_k}$  and a MALA kernel invariant for each intermediate  
   distribution ( $M$  as in Remark 2);  
   // score the step by its worst per-step ESS  
8   Record  $\text{ESS}_{\min}$ , the smallest per-step ESS over the  $M$  steps;  
9   if  $\text{ESS}_{\min} \geq \tau_{\text{SMC}}$  then  
10  | return  $t_k;$   
11  else  
12  | // shrink the step toward  $t_{k-1}$   
    |  $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1});$   
13  end  
14 end
```

---

default SMC floor is zero, which disables this gate and sends the proposed increment directly to trained-flow validation.

When enabled, these SMC runs involve only the potentials  $U_0$  and  $U$ , never the flow map. They use no QT samples, so their per-step ESS can overlook a mode the SMC population never reaches. We therefore use this ESS only to screen an increment; it is a heuristic measure of bridge difficulty, not a coverage certificate. The trained-flow validation in step (v) supplies the acceptance gate, while QT supplies the independent wide-coverage set. By default  $\tau_{\text{SMC}} = 0$ , so Algorithm 3 returns  $t_{\text{init}}$  without running SMC. For a positive floor, it returns the first candidate along the shrink sequence whose estimated ESS clears that floor. The shrink factor is  $\gamma = 0.7$ . As  $t_k$  approaches  $t_{k-1}$ , the bridge becomes trivial and its population ESS approaches 1; in finite samples we also impose the outer loop’s retry and stage limits.

## 4.2 The full training procedure

The two ingredients assemble into a single loop (Algorithm 4) that grows the ladder one stage at a time, from  $U_0$  to  $U$ , carrying a validation population  $\mathcal{Y}_k \approx \mu_k$  of fixed size  $N$  across stages. Stage  $k$  runs five steps. Steps (iii)–(v) are wrapped in an acceptance loop that shrinks the increment and retrains whenever the trained flow does not reach the validation ESS floor  $\tau_v$ . The outer loop stops once  $t_k$  reaches 1, so the number of stages  $K$  is selected automatically.

- (i) *Source population.* Stage  $k$  uses two sample sizes: the *validation size*  $N$  and the *batch size*  $B$ . The complete population  $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\} \approx \mu_{k-1}$  is carried over from the previous stage. A working copy of all  $N$  particles supplies the SMC calculation in step (ii), and all  $N$  particles initialize QT in

step (iii). Neither operation first draws a separately sized subset. Training batches are drawn at run time,  $B$  samples per gradient step, from  $\mathcal{Y}_{k-1}$ ; no separate training set is built.

- (ii) *Adaptive step.* The next level  $t_k$  is fixed by the adaptive selection scheme (Algorithm 3). The first stage uses the safe start  $t_{\text{init}} = t_{\text{safe}} = 0.2$ . Later stages extrapolate the accepted increments,  $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$  with enlarge factor  $\Gamma = 1.5$ ; the step sizes grow, since only the first few stages are difficult. The default  $\tau_{\text{SMC}} = 0$  accepts this initial guess without the optional SMC screen. If a positive SMC floor is supplied, Algorithm 3 instead shrinks from  $t_{\text{init}}$  until the classical SMC ESS on all  $N$  particles clears that floor. This sets the stage target  $\mu_k \propto \exp(-U_{t_k})$  with  $U_{t_k} = (1-t_k)U_0 + t_k U_1$ .
- (iii) *Wide-coverage set.* Quench and temper (Algorithm 2) acts on the complete validation population  $\mathcal{Y}_{k-1}$  under the stage target  $U_{t_k}$ , producing  $\hat{\mathcal{Y}}_k = \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$  with  $N$  samples distributed as  $\hat{\mu}_k$ . It supplies the wide-coverage half of the mixture term in the training loss.
- (iv) *Training loss.* The flow  $G_k$  is trained by  $\text{KL} + \text{X}_{\mu_k} + \text{X}_{(\hat{\mu}_k + \bar{\nu}_k)/2}$ , the balanced loss of Section 3.1 specialized to the stage  $\mu_{k-1} \rightarrow \mu_k$ . With model  $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$  and its stop-gradient copy  $\bar{\nu}_k$ , the stage log-ratio is

$$z_k(y) = U_{t_{k-1}}(G_k(y)) - U_{t_k}(y) - \log |\det J_{G_k}(y)|,$$

and the loss reads

$$\mathcal{L}[G_k] = \underbrace{\mathbb{E}_{y \sim \mu_k} [z_k(y)]}_{\text{KL}(\mu_k \parallel \nu_k)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu_k} [z_k(y) - z_k(y')]}_{\text{X}_{\mu_k}(\mu_k \parallel \nu_k)} + \underbrace{\mathbb{E}_{y, y' \sim (\hat{\mu}_k + \bar{\nu}_k)/2} [z_k(y) - z_k(y')]}_{\text{X}_{(\hat{\mu}_k + \bar{\nu}_k)/2}(\mu_k \parallel \nu_k)}, \quad (18)$$

its  $\mu_k$ -expectations estimated from the biased score-free annealing surrogate: the level- $(k-1)$  batch  $\mathcal{X}$  is pushed to  $\bar{\mathcal{X}}_\nu = G_k^{-1}(\mathcal{X})$ , distributed as  $\nu_k$ , and annealed toward  $\mu_k$  by Algorithm 1 with  $G = G_k$ . The wide-coverage measure  $\hat{\mu}_k$  is realized by the set  $\hat{\mathcal{Y}}_k$  of step (iii).

- (v) *Validation set update.* Once  $G_k$  is trained, each  $y \in \mathcal{Y}_{k-1}$  is pushed through the trained inverse to  $\tilde{y} = G_k^{-1}(y)$ . This gives the proposal population  $\tilde{\mathcal{Y}}_k = G_k^{-1}(\mathcal{Y}_{k-1})$  from  $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$ , together with the importance weights

$$w_k(\tilde{y}) := \exp(z_k(\tilde{y})), \quad \tilde{y} \in \tilde{\mathcal{Y}}_k.$$

The pair  $(\tilde{\mathcal{Y}}_k, w_k)$  provides the final check on the increment. If  $\text{ESS}(w_k) < \tau_v$ , with default  $\tau_v = 0.6$ , we discard the candidate, shrink to  $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ , and repeat steps (iii)–(v) at the reduced step. Otherwise we retain  $G_k$  as the stage map and advance the validation population to  $\mathcal{Y}_k = \text{MALA}(\text{resample}(\tilde{\mathcal{Y}}_k, w_k; N), U_{t_k}) \approx \mu_k$ .

The trained ladder is a Boltzmann generator for  $\mu$ . Starting from  $x \sim \mu_0$ , each stage applies its map  $G_k^{-1}$ , then reweights and resamples the pair  $(\tilde{\mathcal{Y}}_k, w_k)$  from step (v), carrying the samples from source to target one bridge at a time.

## 5 Numerical experiments on model problems

We test the three roles of the log-ratio-variation terms before applying the full objective to an annealed lattice model. The 2D targets isolate accuracy, mode discovery, and leakage control; the product multi-well measures how the accuracy gain scales with dimension; and the tilted  $\phi^4$  theory exposes a collective-barrier failure that ESS alone cannot detect. The  $p$ -state clock model then tests  $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$  along a shared annealing ladder. Throughout, ESS is paired with a diagnostic that detects missing modes: coverage on the multimodal benchmarks, the minority-phase weight on  $\phi^4$ , and sector occupancy on the clock model. All results in this section were rerun with `jflows` v0.5.0.

---

**Algorithm 4:** Adaptive-temperature Boltzmann generator training

---

**Input:** potentials  $U_0, U$ , validation size  $N$ , batch size  $B$ , SMC ESS floor  $\tau_{\text{SMC}} = 0$ , validation ESS floor  $\tau_v = 0.6$ , shrink factor  $\gamma = 0.7$ , enlarge factor  $\Gamma = 1.5$ , safe start  $t_{\text{safe}} = 0.2$

**Output:** ladder  $0 = t_0 < \dots < t_K = 1$  and flow maps  $\{G_k\}_{k=1}^K$  with  $(G_k^{-1})_{\#} \mu_{k-1} \approx \mu_k$

// validation population,  $\mathcal{Y}_k \approx \mu_k$

```
1  $\mathcal{Y}_0 \leftarrow N$  samples from  $\mu_0$ ,  $t_0 \leftarrow 0$ ,  $k \leftarrow 0$ ;  
2 while  $t_k < 1$  do  
3    $k \leftarrow k + 1$ ;  
4   Form the stage source potential,  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U_1$ ;  
   // (i) complete source population  
5   Use all  $N$  particles in  $\mathcal{Y}_{k-1}$  for stage selection, QT, and validation;  
   // (ii) adaptive temperature selection  
6   if  $k = 1$  then  
7     Set  $t_{\text{init}} = t_{\text{safe}}$ ;  
8   else  
9     Set  $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ ;  
10  end  
11  Choose  $t_k$  by adaptive selection (Algorithm 3) on  $\mathcal{Y}_{k-1}$  with initial guess  $t_{\text{init}}$ ;  
12  Form the target potential  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU_1$ ;  
13  repeat  
14    Initialize  $G_k$  to the identity map;  
    // (iii) wide-coverage set  
15    Apply quench and temper to all  $N$  validation particles,  $\hat{\mathcal{Y}}_k \leftarrow \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$  (Algorithm 2);  
    // (iv) balanced log-ratio-variation loss  
16    for each gradient step do  
17      Draw a training batch of  $B$  samples  $\mathcal{X}$  from the validation set  $\mathcal{Y}_{k-1}$ ;  
18      Form the detached model batch,  $\tilde{\mathcal{X}}_\nu \leftarrow G_k^{-1}(\mathcal{X})$ , distributed as  $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$ ;  
19      Form the biased target surrogate  $\mathcal{X}_\mu$  by score-free annealing (Algorithm 1, from  $\nu_k$  toward  $\mu_k$ );  
20      Draw a wide-coverage batch  $\hat{\mathcal{X}}_\mu \subset \hat{\mathcal{Y}}_k$  and rejuvenate,  $\hat{\mathcal{X}}_\mu \leftarrow \text{MALA}(\hat{\mathcal{X}}_\mu, U_{t_k})$ ;  
21      Update  $G_k$  by a gradient step on  $\text{KL} + \text{X}_{\mu_k} + \text{X}_{(\hat{\mu}_k + \bar{\nu}_k)/2}$  in (18), from  $\mathcal{X}_\mu$  and the balanced  
      mixture of  $\hat{\mathcal{X}}_\mu$  and  $\tilde{\mathcal{X}}_\nu$ ;  
22    end  
    // (v) validation set update  
23    Form the trained-flow proposal and weights  $(\tilde{\mathcal{Y}}_k, w_k)$  as in step (v);  
24    if  $\text{ESS}(w_k) < \tau_v$  then  
25      // abort the step and shrink toward  $t_{k-1}$   
       $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ , then re-form  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU_1$ ;  
26    else  
27      Retain the trained stage map  $G_k$ ;  
28      Resample  $N$  particles,  $\mathcal{Y}_k \leftarrow \text{resample}(\tilde{\mathcal{Y}}_k, w_k; N)$ ;  
29      Employ MALA rejuvenation to obtain  $\mathcal{Y}_k \leftarrow \text{MALA}(\mathcal{Y}_k, U_{t_k})$ ;  
30    end  
31  until  $\text{ESS}(w_k) \geq \tau_v$ ;  
32 end  
33  $K \leftarrow k$ ;  
34 return  $\{t_k\}_{k=0}^K$  and  $\{G_k\}_{k=1}^K$ ;
```

---



## 5.1 2D benchmark distributions

We consider four 2D targets: Two-Moon, Three-Well, Himmelblau, and Sparse. Two-Moon has two distorted crescent components and isolates finite-batch accuracy because both components lie within reach of the source. The remaining targets isolate mode collapse: Three-Well contains a low-probability basin, Himmelblau has four separated wells, and Sparse places two tight modes far into the source tails.

For every target, we initialize the same neural spline flow at the identity and compare four objectives:

$$\text{KL}, \quad \text{KL}+X_\mu, \quad \text{KL}+X_\mu+X_{\hat{\mu}}, \quad \text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}.$$

The flow has 32 spline bins, 6 transforms, and conditioner widths (128,128). Each training step constructs a target surrogate by a single-stage AIS update through the current flow. The rejuvenation kernel uses 100 MALA steps of size  $10^{-3}$  for Two-Moon and 50 MALA steps of size  $2 \times 10^{-3}$  for the other targets. For both KLXX objectives, quench and temper acts on the complete fixed validation population, with melt scale 2 and an Armijo L-BFGS quench using initial step 0.5 and 200 iterations. The final ESS is computed from flow importance weights on that full population. Coverage, with  $\kappa = 5$  against an independently generated quench-and-temper reference set, measures the fraction of reference configurations represented by the pushforward through the nearest-neighbor test of Section 3.4. Hence a collapsed flow can have ESS near one but cannot have full coverage.

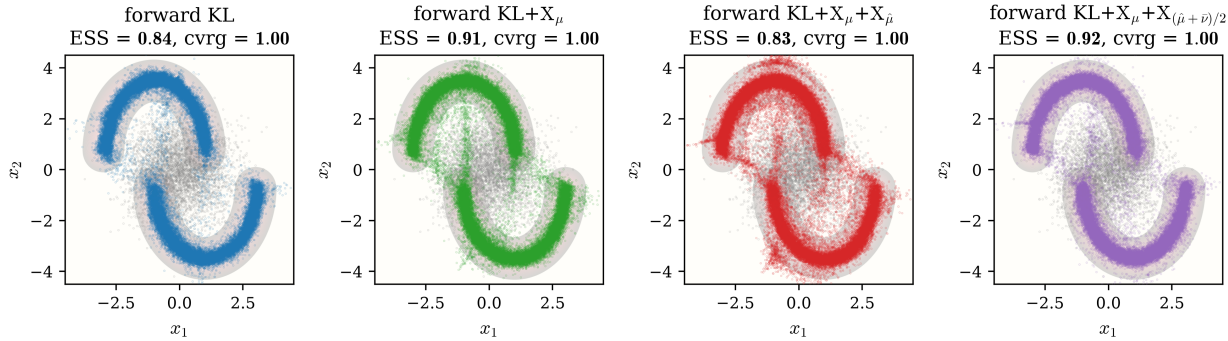


Figure 1: Two-Moon target. Columns, from left to right, are forward KL,  $\text{KL}+X_\mu$ ,  $\text{KL}+X_\mu+X_{\hat{\mu}}$ , and  $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$ . Each panel overlays the learned pushforward on the target energy and reports its final ESS and coverage. Gray points show the source.

**Two-Moon.** The source is  $\mathcal{N}(0, I_2)$  and both crescents lie within its reach, so all four losses attain coverage 1. With batch size 500, 200 Adam steps, and learning rate  $10^{-3}$ , the bare forward KL reaches ESS 0.8358. Adding  $X_\mu$  raises it to 0.9109 without additional flow evaluations, while the equal-weight mixture is best at 0.9236 (Figure 1). This benchmark isolates the accuracy role of  $X_\mu$ : the pairwise term constrains the spread of the log-ratio left uncontrolled by the mean-only forward KL objective.

**Three-Well.** The source is  $\mathcal{N}(0, 0.25^2 I_2)$ , narrower than the well separation and initially covering no basin. Training uses batch size 1000, 500 steps, and learning rate  $10^{-3}$ . Forward KL and  $\text{KL}+X_\mu$  fill the two favored wells and report ESS 0.9884 and 0.9902, respectively, while their coverage remains 0.7595. Quench and temper discovers the suppressed third well. The  $\hat{\mu}$  and equal-weight mixtures cover all three wells in the sample panels, with scalar coverages 0.9783 and 0.9768 and ESS 0.9611 and 0.9893 (Figure 2).

**Himmelblau.** The source is  $\mathcal{N}(0, I_2)$ , whereas the four wells lie at radius approximately 3–4. Training uses batch size 1000, 1000 steps, and learning rate  $10^{-3}$ . Forward KL and  $\text{KL}+X_\mu$  capture only two wells, giving coverage 0.5160 despite ESS 0.9316 and 0.9809. Both discovery mixtures attain coverage 1. The  $\hat{\mu}$ -only flow has ESS 0.9562, whereas adding the  $\bar{\nu}$  half suppresses intermodal overcoverage and raises the ESS to 0.9736 (Figure 3).

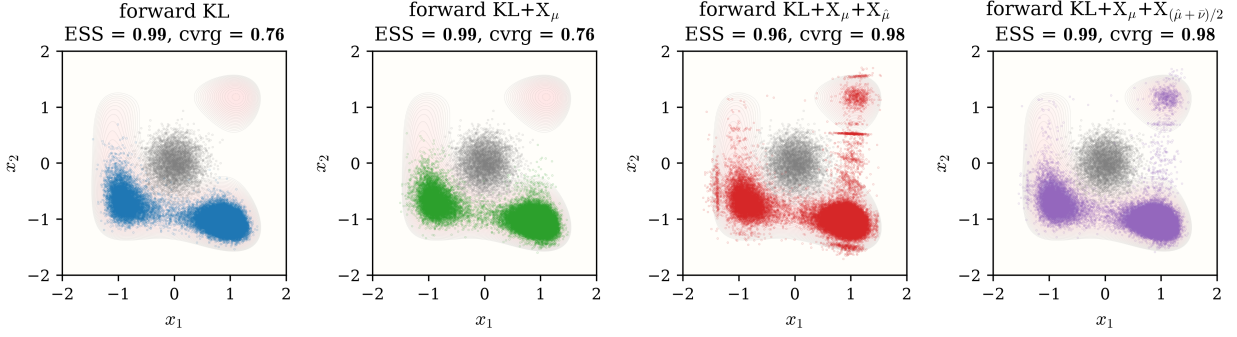


Figure 2: Three-Well target. Methods and annotations are as in Figure 1.

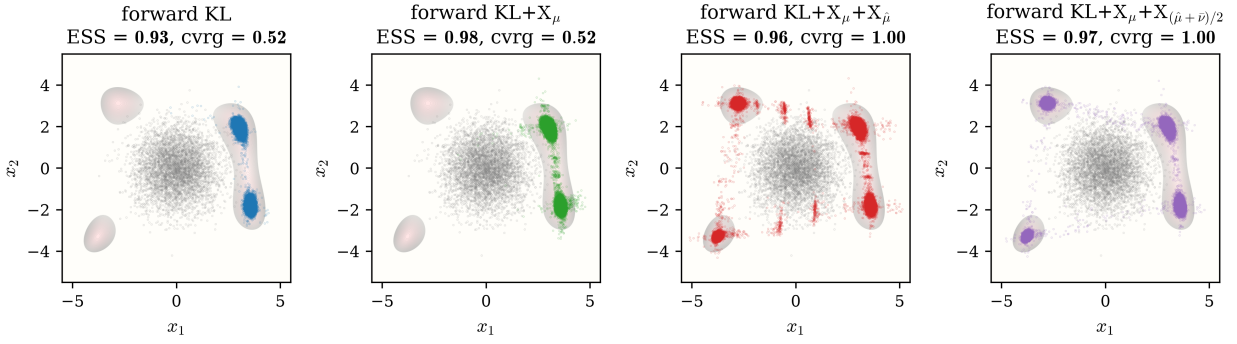


Figure 3: Himmelblau target. Methods and annotations are as in Figure 1.

**Sparse.** The target is a four-component Gaussian mixture with component standard deviation 0.08: two modes lie near the source center and two lie in opposite corners, more than five source standard deviations away. With source standard deviation 0.6, batch size 1000, 1000 steps, and learning rate  $5 \times 10^{-3}$ , forward KL and KL+ $X_\mu$  cover only the central pair. Their coverage is 0.5335, while their ESS is 0.9116 and 0.9266. The  $\hat{\mu}$ -only mixture restores coverage but pays a leakage cost, with ESS 0.8262. The equal-weight mixture also attains coverage 1 and improves on every other objective, reaching ESS 0.9555.

| target     | KL            | KL+ $X_\mu$   | KL+ $X_\mu$ + $X_{\hat{\mu}}$ | KL+ $X_\mu$ + $X_{(\hat{\mu}+\bar{\nu})/2}$ |
|------------|---------------|---------------|-------------------------------|---------------------------------------------|
| Two-Moon   | 0.836 (1.000) | 0.911 (1.000) | 0.835 (1.000)                 | <b>0.924</b> (1.000)                        |
| Three-Well | 0.988 (0.760) | 0.990 (0.760) | 0.961 (0.978)                 | <b>0.989</b> (0.977)                        |
| Himmelblau | 0.932 (0.516) | 0.981 (0.516) | 0.956 (1.000)                 | <b>0.974</b> (1.000)                        |
| Sparse     | 0.912 (0.534) | 0.927 (0.534) | 0.826 (1.000)                 | <b>0.956</b> (1.000)                        |

Table 2: Final ESS, with  $\kappa = 5$  coverage in parentheses, on the four 2D targets. Bold marks the highest ESS among methods whose sample panel covers every target mode. The scalar coverage diagnostic slightly undercounts the visibly occupied third mode on Three-Well.

The four targets separate the objective’s three mechanisms. The  $X_\mu$  term improves finite-batch accuracy when discovery is not limiting. The  $X_{\hat{\mu}}$  term reveals modes absent from the live AIS surrogate. The  $\bar{\nu}$  half then penalizes the low-probability bridges introduced by aggressive discovery. The validation ESS of  $X_{(\hat{\mu}+\bar{\nu})/2}$  is often close to that of  $X_{\hat{\mu}}$ , but its samples are visibly cleaner and avoid overcovering intermodal regions. In this rerun, the equal-weight mixture combines all three effects and has the highest mode-complete ESS on every target in Table 2.

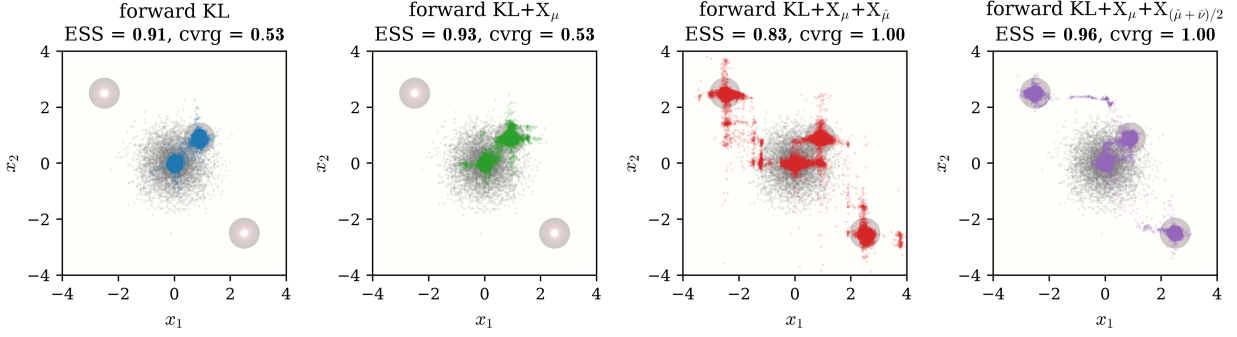


Figure 4: Sparse target. Methods and annotations are as in Figure 1.

## 5.2 High-dimensional product multi-well

We next isolate accuracy scaling over the sixteen dimensions  $d = 16, 32, \dots, 256$ . Letting  $k = \lfloor \log_2 d \rfloor$ , we use

$$U(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|^2 + 12 \sum_{i=1}^k \exp(-x_i^2). \quad (19)$$

The first  $k$  coordinates are symmetric double wells with minima at  $\pm\sqrt{\ln 24}$  and barrier approximately 9.9, while the remaining  $d-k$  coordinates are standard Gaussian. The target therefore has exactly  $2^k$  equal-weight modes, where  $2^k \leq d < 2^{k+1}$ . Multimodality occupies only a  $k$ -dimensional subspace, but the log-weight error accumulates across all  $d$  coordinates.

All four objectives use the same identity-initialized neural spline flow on  $[-4, 4]^d$ , with 16 bins, 6 transforms, and conditioner widths (256, 256). At every dimension, training uses batch size 250, 2000 Adam steps at learning rate  $10^{-3}$ , gradient clipping at  $10^2$ , and a fixed validation population of size  $5000d$ . The live target surrogate uses a single AIS hop followed by 50 MALA steps of size  $2 \times 10^{-3}$ . Quench and temper uses  $1000d$  particles, melt scale 2, and an Armijo L-BFGS quench with initial step 0.5 and 200 iterations. Final validation ESS is evaluated on  $\max(200d, 80000)$  fresh source samples.

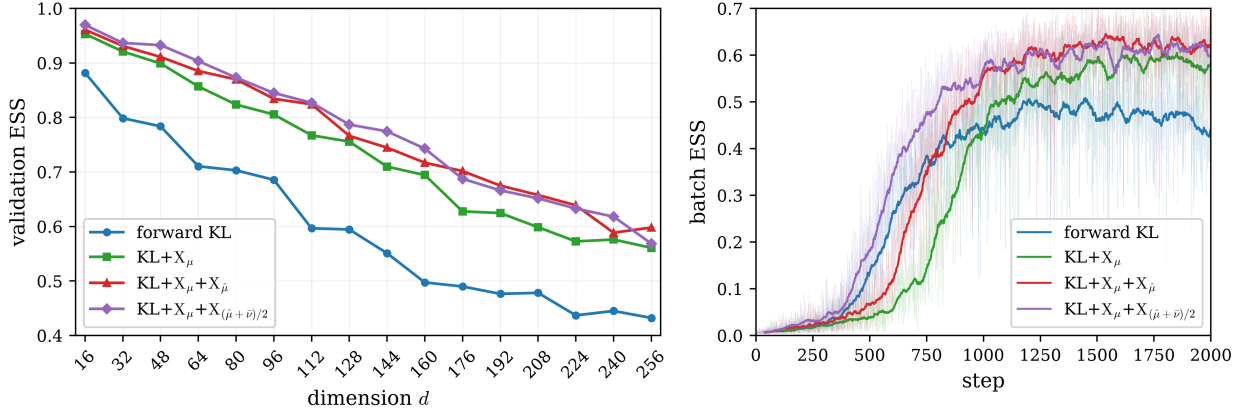


Figure 5: ESS on the product multi-well target (19). Left: final validation ESS over all sixteen tested dimensions. Right: batch ESS during the  $d = 256$  training run; faint curves are raw values and bold curves are 40-step moving averages.

Figure 5 shows a persistent advantage from X-regularization. Every X-regularized objective has higher validation ESS than bare forward KL at all sixteen dimensions. At  $d = 256$ , the final values are 0.4322 for forward KL, 0.5608 for  $\text{KL}+X_\mu$ , 0.5979 for the  $\hat{\mu}$  mixture, and 0.5687 for the equal-weight mixture. The two KLXX objectives behave very similarly across dimension, while all three X-regularized losses continue

improving after forward KL begins to plateau in the  $d = 256$  training history. Thus the scaling advantage is not obtained by sacrificing the training trajectory.

### 5.3 Lattice $\phi^4$ field theory: fake ESS at a collective barrier

The tilted scalar  $\phi^4$  theory provides a sharp fake-ESS test. On a periodic  $L \times L$  lattice, with  $d = L^2$  and magnetization  $m(\phi) = d^{-1} \sum_j \phi_j$ , the action is

$$U(\phi) = -0.8 \sum_{\langle jl \rangle} \phi_j \phi_l + \sum_{j=1}^d \left[ \phi_j^2 + \frac{1}{2} (\phi_j^2 - 1)^2 \right] + h \sum_{j=1}^d \phi_j.$$

We take  $h = 0.0257$  at  $L = 6$  and  $h = 0.0144$  at  $L = 8$ . This size-dependent field keeps the extensive tilt scale nearly fixed:  $hL^2 = 0.9252$  at  $L = 6$  and  $hL^2 = 0.9216$  at  $L = 8$ . Thus the two targets have comparable phase imbalance despite their different volumes. In the broken phase the field orders near two vacua  $\pm \mathbf{v}_1$ , separated by a collective domain-wall barrier. The positive linear term gives the two phases unequal probability, so the minority weight  $p_+ = \mathbb{P}(m > 0)$  is not fixed by symmetry and must be learned.

Parallel tempering gives the references  $p_+ = 0.1251 \pm 0.0009$  and  $0.1284 \pm 0.0014$ , statistically consistent with occupancies 0.1260 and 0.1268 in the independent one-million-sample resampled symmetry-confined MALA traces. Each loss is run at seeds 0, 1, 2 using an identity-initialized neural spline flow on  $[-3, 3]^d$ , with 16 bins, 6 transforms, and conditioner widths (256, 256). The source is  $\mathcal{N}(0, 0.25I_d)$ , centered on the barrier. Training uses a fixed validation population of  $10^5$  source samples, batch size 500, 2000 Adam steps at learning rate  $10^{-3}$ , and gradient clipping at  $10^3$ . The live surrogate uses one AIS hop and 50 MALA steps of size  $2 \times 10^{-3}$ . For both KLXX objectives, quench and temper acts on the complete validation population, with melt scale 2 and an Armijo L-BFGS quench using initial step 0.1 and 200 iterations.

|                                             | $L = 6$ ( $d = 36$ , $h = 0.0257$ ) |                      | $L = 8$ ( $d = 64$ , $h = 0.0144$ ) |                      |
|---------------------------------------------|-------------------------------------|----------------------|-------------------------------------|----------------------|
|                                             | ESS                                 | $p_+$                | ESS                                 | $p_+$                |
| forward KL                                  | 0.794/0.840/0.826                   | 1/1/0                | 0.622/0.686/0.684                   | 1/0/1                |
| KL+ $X_\mu$                                 | 0.935/0.933/0.945                   | 1/1/0                | 0.804/0.824/0.821                   | 1/0/0                |
| KL+ $X_\mu$ + $X_{\hat{\mu}}$               | 0.875/0.873/0.871                   | 0.1273/0.1275/0.1272 | 0.636/0.645/0.654                   | 0.1251/0.1238/0.1242 |
| KL+ $X_\mu$ + $X_{(\hat{\mu}+\bar{\nu})/2}$ | <b>0.896/0.877/0.894</b>            | 0.1274/0.1277/0.1273 | <b>0.690/0.676/0.681</b>            | 0.1312/0.1295/0.1293 |
| PT reference                                | —                                   | $0.1251 \pm 0.0009$  | —                                   | $0.1284 \pm 0.0014$  |

Table 3: Final ESS and reweighted minority-phase weight  $p_+$  for three seeds on the two tilted  $\phi^4$  targets,  $(L, h) = (6, 0.0257)$  and  $(8, 0.0144)$ . The size-dependent field keeps  $hL^2 \simeq 0.923$ , making the phase imbalance comparable across the two volumes. Integer values 0 and 1 indicate collapse onto one vacuum; bold marks the higher ESS among the objectives that recover both vacua.

Forward KL and KL+ $X_\mu$  collapse at every size and seed. Because the AIS chain starts from the flow pushforward and cannot cross the collective barrier, the second vacuum remains absent and reweighting gives  $p_+ = 0$  or 1. Nevertheless, KL+ $X_\mu$  has the highest ESS at both sizes: sharpening the captured vacuum improves calibration on the reached support while making the global failure no less severe.

Both quench-and-temper objectives populate both vacua at every seed. Their recovered  $p_+$  lies between 0.1238 and 0.1312, within 0.005 of both the parallel-tempering estimate and the saved resampled mirror-MALA reference at the corresponding lattice size. This recovery has an unavoidable ESS cost: a diffeomorphism maps the connected Gaussian source to a connected pushforward, so covering both vacua forces a low-weight bridge across the suppressed inter-phase region. At  $L = 8$ , the equal-weight mixture recovers part of this cost, with ESS 0.6756–0.6902 rather than 0.6355–0.6541 for the  $\hat{\mu}$ -only objective. The  $\bar{\nu}$  component penalizes precisely the leaked bridge mass while preserving discovery.

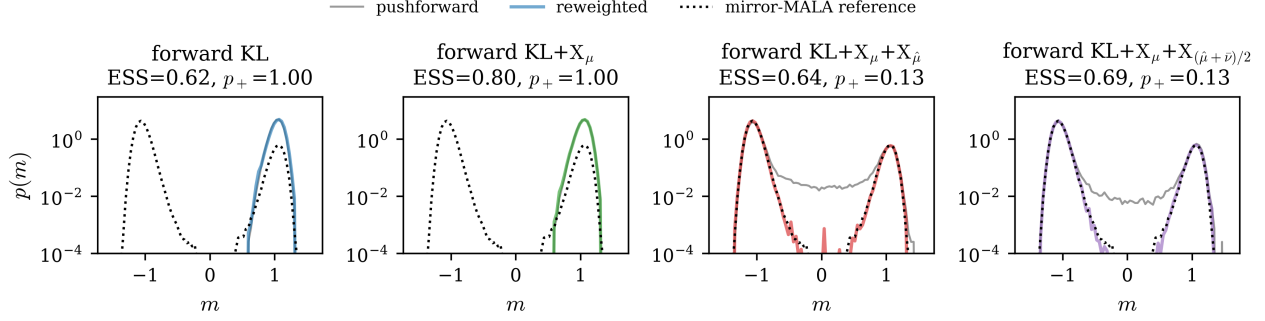


Figure 6: Magnetization density on the  $L = 8$  lattice at seed 0. Each panel shows the pushforward density (gray), the reweighted density (color), and the saved resampled mirror-MALA reference density (dotted); the title reports ESS and the reweighted  $p_+$ .

#### 5.4 Boltzmann generator on $p$ -state clock model

The preceding benchmarks use a single flow. We now test the full annealed construction on the  $p = 6$  clock model on a periodic  $8 \times 8$  lattice,

$$U(\boldsymbol{\theta}) = - \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) - \frac{1}{2} \sum_{j=1}^d \cos(6\theta_j), \quad \boldsymbol{\theta} \in [-\pi, \pi)^d, \quad d = 64, \quad (20)$$

where  $\langle jl \rangle$  runs over nearest-neighbor pairs. The source is uniform on the torus and the bridge is  $U_t = tU$ . At low temperature, the target separates into six symmetry-related clock sectors.

Each stage uses an identity-initialized neural circular spline flow with 16 bins, 6 autoregressive transforms, and conditioner widths (256, 256). The validation set contains  $4 \times 10^5$  particles, and both SMC and temper act directly on this complete population. Every stage uses 3000 Adam steps at learning rate  $10^{-3}$ , a four-stage training AIS surrogate with 100 MALA steps of size  $10^{-3}$ , energy clipping at  $10^3$ , and gradient clipping at  $10^2$ . We report batch sizes  $B = 2000, 1000, 500, 250$ .

The comparison is schedule-matched. Forward KL first constructs an adaptive ladder using SMC preselection and an ESS acceptance gate. The loss  $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$  is then trained on the identical accepted values of  $t$ , without a second gate. Hence the two methods see the same bridge increments within each batch-size pair. Every configuration is a single run, so individual stage values carry single-run fluctuation.

The principal metric is the trained-flow validation ESS of each accepted increment. We define the

$$\text{propagation factor} \quad F := \prod_k \text{ESS}_k^{-1}$$

as an estimate of the factor by which staged reweighting enlarges Monte Carlo error across the full ladder. Smaller  $F$  is better. Table 4 reports every accepted stage and the resulting  $F$ .

The balanced loss raises the geometric mean stage ESS at every reported batch size: from 0.556 to 0.665 at  $B = 2000$ , from 0.549 to 0.643 at  $B = 1000$ , from 0.487 to 0.571 at  $B = 500$ , and from 0.468 to 0.553 at  $B = 250$ . Correspondingly, it reduces  $F$  by factors between 2.93 and 3.82. The gain is not confined to the small-batch regime: at the largest batch,  $F$  falls from 33.9 to 11.6.

The comparison is also consistent level by level. The balanced loss has higher validation ESS on all 28 shared increments, with a particularly clear advantage in the large-batch runs. As the batch shrinks, the adaptive forward-KL schedule lengthens from six to eight reported levels and both propagation factors grow. The X-regularized objective nevertheless loses less ESS before the final target is reached.

The single-site marginal exhibits all six clock wells with nearly equal peak heights. Pair differences are most concentrated at zero for nearest neighbors; as the lattice separation increases, the alignment peak falls and the shoulders near  $\pm\pi/3$  gain mass. This is the expected decay of local angular order with distance and confirms that the staged sampler is not confined to a single global sector.

To quantify whether staged resampling preserves the sixfold symmetry, we rebuild fresh samples through the trained  $B = 2000$  ladders at sizes  $N = 10^4 2^k$ ,  $k = 0, \dots, 6$ , using equal total work of  $2.56 \times 10^6$  particles

| stage $k$                                     | 1            | 2            | 3            | 4            | 5            | 6            | 7            | 8            | $F$          |
|-----------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| $t_k$ ( $B = 2000$ )                          | 0.250        | 0.495        | 0.663        | 0.779        | 0.887        | 1.000        | ×            | ×            |              |
| forward KL                                    | 0.751        | 0.570        | 0.471        | 0.474        | 0.511        | 0.602        | —            | —            | 33.9         |
| KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$ | <b>0.921</b> | <b>0.759</b> | <b>0.608</b> | <b>0.562</b> | <b>0.558</b> | <b>0.649</b> | —            | —            | <b>11.6</b>  |
| $t_k$ ( $B = 1000$ )                          | 0.250        | 0.495        | 0.613        | 0.728        | 0.841        | 0.952        | 1.000        | ×            |              |
| forward KL                                    | 0.672        | 0.479        | 0.580        | 0.463        | 0.427        | 0.520        | 0.789        | —            | 66.2         |
| KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$ | <b>0.866</b> | <b>0.642</b> | <b>0.700</b> | <b>0.545</b> | <b>0.462</b> | <b>0.541</b> | <b>0.859</b> | —            | <b>22.0</b>  |
| $t_k$ ( $B = 500$ )                           | 0.250        | 0.421        | 0.590        | 0.705        | 0.784        | 0.895        | 1.000        | ×            |              |
| forward KL                                    | 0.580        | 0.549        | 0.408        | 0.432        | 0.525        | 0.417        | 0.530        | —            | 153.5        |
| KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$ | <b>0.778</b> | <b>0.695</b> | <b>0.506</b> | <b>0.500</b> | <b>0.595</b> | <b>0.429</b> | <b>0.566</b> | —            | <b>50.7</b>  |
| $t_k$ ( $B = 250$ )                           | 0.250        | 0.421        | 0.539        | 0.654        | 0.734        | 0.811        | 0.887        | 1.000        |              |
| forward KL                                    | 0.468        | 0.448        | 0.495        | 0.428        | 0.506        | 0.484        | 0.498        | 0.424        | 435.1        |
| KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$ | <b>0.659</b> | <b>0.586</b> | <b>0.601</b> | <b>0.480</b> | <b>0.563</b> | <b>0.536</b> | <b>0.564</b> | <b>0.464</b> | <b>113.8</b> |

Table 4: Trained-flow validation ESS on every accepted stage of the schedule-matched clock-model comparison between forward KL and KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$ . Within each batch-size block, both losses traverse the same ladder. The final column is  $F = \prod_k \text{ESS}_k^{-1}$ ; bold marks the better loss at each stage and the smaller  $F$ .

per row. We define

$$\text{occupancy bias} := \frac{1}{6} \sum_{s=0}^5 \left| p_s - \frac{1}{6} \right|.$$

Both methods follow the Monte Carlo rate: the log-log slopes of occupancy bias against  $N$  are  $-0.459$  for forward KL and  $-0.479$  for the balanced loss. This excludes a detectable particle-count-independent sector imbalance in either staged sampler over the tested range. The balanced curve is lower at all seven reported sizes, reducing the mean bias from 0.03735 to 0.02397 at  $N = 10^4$  and from 0.00552 to 0.00358 at  $N = 6.4 \times 10^5$ . Every reported rebuild retains all six sectors. Together with the six peaks in Figure 7, the scaling test shows that the staged samples are not confined to one global sector. Because each flow ladder is one training run, the amplitude difference is evidence for these  $B = 2000$  realizations rather than an estimate of variation over independently trained models.

## 6 Boltzmann generators on molecular potential energies

The model problems above isolate accuracy, discovery, and collective barriers on targets whose measure is unambiguous. The application that motivates a Boltzmann generator is a molecule. Its equilibrium conformers determine thermodynamic averages, free-energy differences, and the structural ensembles used downstream in computational chemistry (Noé et al., 2019). Conventional molecular dynamics provides local physical trajectories, but torsional barriers can make transitions between conformers rare. A trained invertible map instead proposes independent configurations together with a tractable density, after which ordinary importance reweighting corrects the residual error. The payoff is therefore not an unweighted picture of plausible conformers, but a reusable proposal whose statistical error can be measured.

A molecular target exercises the full construction at once. The Cartesian force field contains a steep nonbonded core; removing rigid translation and rotation produces a mixed Euclidean-periodic internal-coordinate domain; and the physically intended support may contain only one stereochemical component. These are target-definition questions, not implementation details: changing the solvent model, omitting the coordinate Jacobian, wrapping a bond coordinate, or mixing mirror components changes the probability measure being learned. This section consequently fixes the physical model, coordinate measure, and support before reporting a molecular result. It defines three implicit-solvent benchmarks—glycerol, neutral diethanolamine, and L-alanine dipeptide—and the adaptive mixed-domain training protocol that will be used for all three.



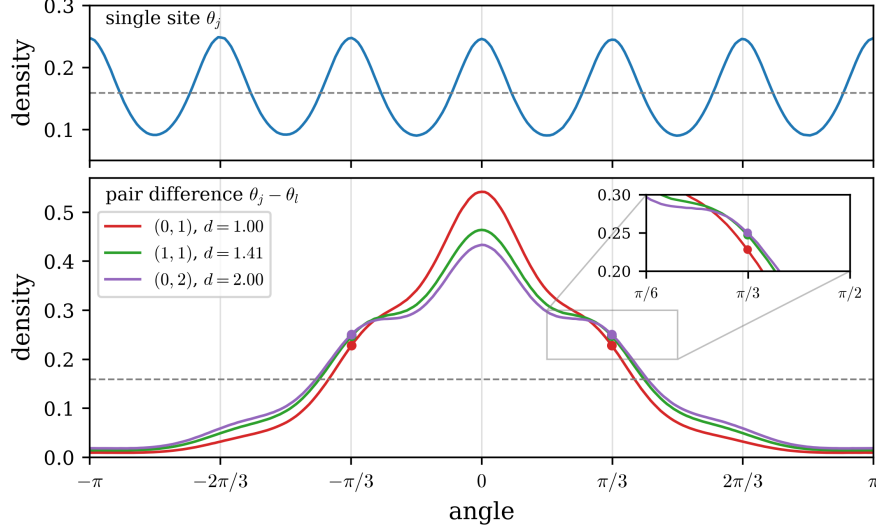


Figure 7: Angle densities from  $2 \times 10^6$  fresh samples advanced through the trained  $B = 2000$   $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{v})/2}$  ladder by map, reweighting, resampling, and MALA. The upper panel is the single-site marginal. The lower panel gives pair differences at offsets (0,1), (1,1), and (0,2); dots and the inset resolve the one-clock-step angle  $\pi/3$ . Dashed lines show the uniform density.

The molecular production runs are not yet complete. The discussion below is deliberately divided into what has already been fixed—Hamiltonians, coordinate charts, reference data, and controller semantics—and what remains empirical. No interrupted run, compile smoke test, or softened-potential diagnostic is promoted as a Boltzmann-generator result.

## 6.1 Bundle-defined implicit-solvent targets

For Cartesian coordinates  $x$ , let  $E(x)$  denote the complete molecular-mechanics energy

$$E(x) = E_{\text{bond}} + E_{\text{angle}} + E_{\text{torsion}} + E_{\text{LJ}} + E_{\text{Coul}} + E_{\text{GB}}^{\text{OBC1}} + E_{\text{ACE}}. \quad (21)$$

The bonded and vacuum nonbonded part has the familiar Amber form (Cornell et al., 1995)

$$E_{\text{vac}}(x) = \sum_b k_b (r_b - r_b^0)^2 + \sum_a k_a (\theta_a - \theta_a^0)^2 + \sum_\tau \frac{V_\tau}{2} [1 + \cos(n_\tau \varphi_\tau - \gamma_\tau)] \\ + \sum_{\substack{i < j \\ \text{nonbonded}}} \left\{ 4\epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + k_e \frac{q_i q_j}{r_{ij}} \right\}, \quad (22)$$

with the topology fixing exclusions and the usual scaled interactions. OBC1 adds a generalized-Born approximation to electrostatic solvation, and ACE adds its nonpolar contribution. Thus the benchmark is not a vacuum force field with a post-processing correction: all terms in (21) belong to the energy used by sampling and reweighting.

At temperature  $T$ , the Cartesian target is proportional to  $\exp[-\beta E(x)]$ , with  $\beta = (k_B T)^{-1}$ . Alanine dipeptide uses the 22-atom ACE-ALA-NME Hamiltonian of the FAB benchmark (Midgley et al.): Amber ff96, OBC1 with `igb = 2` and `mbondi2` radii, ACE nonpolar solvation, solute and solvent dielectric constants 1 and 78.5, zero salt, no cutoff, no constraints, and  $T = 300$  K. In particular, the similarly named OpenMM `amber96_obc.xml` model is not substituted, because it uses a different OBC convention. Glycerol and neutral diethanolamine use GAFF2 bonded and Lennard-Jones parameters, AM1-BCC charges, and the same OBC1/ACE solvent convention at 300 K.

A PDB file does not determine this target: it supplies atoms and coordinates but not a unique protonation state, force field, partial charges, solvent model, exclusions, or coordinate chart. Reconstructing those

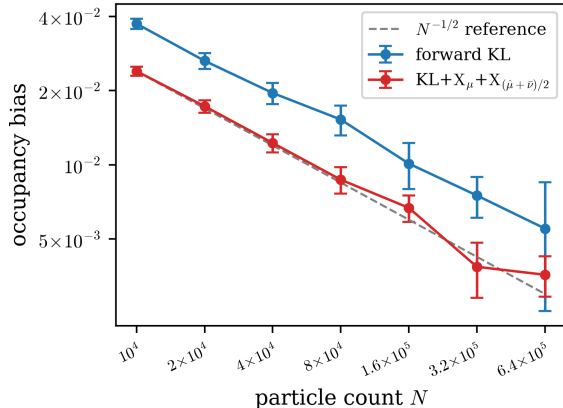


Figure 8: Clock-sector occupancy bias versus fresh rebuild size for the trained  $B = 2000$  ladders. Every row uses equal total work, with  $N = 10^4 2^k$  particles in each of  $2^{8-k}$  independent rebuilds. Error bars are  $\pm 2$  standard errors. The dashed line is an  $N^{-1/2}$  reference anchored at the smallest  $KL+X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$  point.

choices from a filename would make two nominally identical runs capable of learning different measures. Each benchmark is therefore defined by an immutable bundle containing the topology, serialized Hamiltonian, reference coordinates, temperature and units, atom map, internal-coordinate construction, support convention, provenance, and content hashes. OpenMM constructs and independently evaluates the Cartesian system; training uses the corresponding pure-JAX energy frozen in the bundle. Energies and forces are compared against OpenMM before a bundle is admitted. This separation makes the production target reproducible without putting OpenMM inside the differentiated training graph.

## 6.2 Mixed internal coordinates and stereochemical support

For a nonlinear molecule with  $N$  atoms, the Cartesian energy is invariant under three translations and three rotations. Carrying those six null directions through the flow wastes capacity and gives no normalizable laboratory-frame target, so we remove them exactly and retain  $d = 3N - 6$  internal degrees of freedom. A rooted internal-coordinate tree reconstructs  $x = x(q)$  from bond lengths, bond angles, and torsions. The tree, root frame, atom order, and reconstruction convention are frozen in the bundle; they are part of the density transformation rather than plotting metadata.

The coordinate types cannot be treated uniformly. Bond and angle coordinates are whitened with fixed location and scale parameters obtained from the bundle construction, producing unbounded Euclidean variables. Torsions remain angles modulo  $2\pi$ . Thus the molecular domain is not a periodic box but

$$\mathcal{Q} = \mathbb{R}^{d_E} \times \mathbb{T}^{d_T}, \quad d_E + d_T = 3N - 6. \quad (23)$$

The reduced potential with respect to Lebesgue measure on the Euclidean coordinates and Haar measure on the torus is

$$U(q) = \beta E(x(q)) - \log J(q), \quad (24)$$

where  $J(q)$  is the exact Cartesian volume Jacobian of the internal-coordinate reconstruction, including the whitening transforms. Omitting this term would sample  $\exp[-\beta E(x(q))]$  with respect to the wrong base measure and would bias even a perfectly optimized flow. The source is standard normal on  $\mathbb{R}^{d_E}$  and uniform on  $\mathbb{T}^{d_T}$ . The mixed neural spline flow uses ordinary rational-quadratic splines on Euclidean coordinates and circular rational-quadratic splines on torsions. Its masks may couple the two blocks through their conditioners, but its scalar maps respect the topology of each coordinate; bonds are never wrapped and torsions never acquire artificial boundaries.

The support convention is also part of the target. A classical force field can assign the same energy to mirror-related configurations even when the intended molecular benchmark contains a named stereoisomer. If both components are left in the state space, an invertible continuous flow must either learn a physically unwanted mirror mode or place connecting mass across a chirality-changing set. If one component is removed

| target                 | atoms / $d$ | frozen Hamiltonian       | internal domain and support                                            | reference                             |           |
|------------------------|-------------|--------------------------|------------------------------------------------------------------------|---------------------------------------|-----------|
| L-alanine dipeptide    | 22/60       | ff96 + OBC1/ACE          | $\mathbb{R}^{42} \times \mathbb{T}^{18}$ , L component only            | FAB 10 <sup>6</sup> -frame REMD split |           |
| glycerol               | 14/36       | GAFF2/AM1-BCC + OBC1/ACE | $\mathbb{R}^{25} \times \mathbb{T}^{11}$ , full labeled parity support | matched                               | reference |
| neutral diethanolamine | 18/48       | GAFF2/AM1-BCC + OBC1/ACE | $\mathbb{R}^{33} \times \mathbb{T}^{15}$ , both N-pyramid signs        | pending                               | reference |

Table 5: Frozen definitions of the three molecular benchmarks. This is a target-specification table, not a training-result table.

by an after-the-fact filter, the stated flow density is no longer the proposal density used for importance sampling.

The million-frame FAB trajectory lies entirely in the L-alanine component. We therefore encode a smooth L-only chart for alanine dipeptide from the outset rather than adding a soft chirality wall to its energy. Glycerol has no configurational stereocenter, and neutral diethanolamine must retain both signs of its nitrogen-pyramidal coordinate; imposing the alanine half-chart on either would change its distribution. Signed-volume invariants test these conventions on reconstructed configurations, and the accepted sign rule is frozen in the same bundle as the Hamiltonian. Chirality is thereby handled as support geometry, not as an energy-sharpening device.

### 6.3 Exact target, diagnostic regularization, and mixed-domain sampling

The Lennard–Jones core in (22) grows like  $r^{-12}$ . This singularity is physically useful—it excludes atomic overlap—but it creates a severe optimization problem when a smooth map is still assigning appreciable mass to clashes. A few configurations can dominate both the energy and its gradient before the map has learned the broad conformational structure. The remedy must distinguish numerical training protection from a change of target.

The physical potential in (24) is never energy-clipped. In particular, the optimizer screen  $e_{\text{clip}}$  only drops extreme samples from a stochastic loss; it does not alter MALA, SMC weights, validation weights, final importance weights, or the definition of  $U$ . Gradient clipping similarly protects the optimizer update without modifying the target score used by the Monte Carlo kernels. This distinction matters because a finite training loss is not evidence for a finite, normalized, or physically correct surrogate.

The present molecular diagnostic uses an explicit family of regularized Cartesian energies to determine whether a smoother starting target preserves the physical torsional modes. Let  $E_{\text{ref}}$  be the bundle reference energy,  $d(x) = E(x) - E_{\text{ref}}$ ,  $c$  an excess-energy cutoff,  $s > 0$  a logarithmic scale, and  $0 \leq \rho \leq 1$  a linear-tail fraction. Define

$$\mathcal{R}_{c,s,\rho}(d) = \begin{cases} d, & d \leq c, \\ c + (1 - \rho)s \log\left(1 + \frac{d - c}{s}\right) + \rho(d - c), & d > c, \end{cases} \quad (25)$$

and

$$U_{c,s,\rho}(q) = \beta[E_{\text{ref}} + \mathcal{R}_{c,s,\rho}(d(x(q)))] - \log J(q). \quad (26)$$

Only the Cartesian energy is deformed; the coordinate Jacobian and stereochemical support remain exact. The construction is continuously differentiable at  $d = c$ : the tail begins with the same value and unit slope as the physical excess energy. The parameter  $c$  fixes where the deformation starts,  $s$  controls the logarithmic transition, and a positive  $\rho$  retains a coercive linear tail. The endpoint of any accepted continuation is the undeformed potential  $U$ , not a capped substitute. Consequently, a regularized run may diagnose a viable starting landscape, but it cannot by itself serve as the reported molecular equilibrium distribution.

MALA updates the Euclidean coordinates without wrapping and the torsions modulo  $2\pi$ ; its proposal density and accept/reject step use the same mixed-domain convention. Potential-space SMC selects intermediate bridges from honest incremental ESS, resamples, and rejuvenates at the actual intermediate potential.

This is the classical sampling path used to construct and validate a bridge. The flow-proposal AIS surrogate used inside training is deliberately score-free in the flow: the molecular score comes only from the target potential, while the map enters through proposal samples and density ratios.

The adaptive training controller evaluates the final trained proposal on the complete validation set. Its full-validation ESS is the sole post-training stage-acceptance quantity. Per-step batch ESS remains the optimizer-loop monitor; update counts, kept fractions, and compilation diagnostics never shrink or accept a stage. This separation is essential on molecules, where a stable optimizer trace can coexist with a proposal that has not improved the held-out importance weights.

A completed two-seed glycerol cutoff scan shows why the distinction between diagnostic and result is necessary. Above  $c = 100$  kJ/mol, the selected one-dimensional torsion marginals change negligibly on the path to the exact potential, but the two exact-potential endpoint populations still disagree in joint rotamer occupancy. High MALA acceptance therefore did not establish intermode mixing. The scan identifies  $c = 100$  kJ/mol as a candidate starting surrogate, not an equilibrium reference. Production training waits for a full-particle, independently mixed reference diagnostic with a coercive tail and agreement in the joint torsional statistics.

## 6.4 Training and evaluation protocol

The molecular generator uses the same adaptive-temperature construction as the clock experiment, with a mixed-domain spline flow in place of the toroidal flow. Starting from the Gaussian-uniform source, the controller advances a bridge only as far as the SMC preselection gate permits. A candidate flow is then trained for that fixed increment. At every accepted level  $t_k$ , the final trained proposal is evaluated on all validation particles, and its full-set ESS supplies the particle update. The controller therefore cannot manufacture progress from a minibatch fluctuation.

The saved history distinguishes the quantities that serve different roles:  $t_k$ , the per-step batch ESS seen during optimization, and the trained full-validation ESS. Every stage flow, including the most recent rejected attempt, is checkpointed. These checkpoints make it possible to locate a failed bridge or replot a diagnostic without rerunning earlier molecular dynamics, while leaving the accept/reject rule unchanged. The planned comparison is forward KL against  $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$ , using the same accepted schedule within each paired comparison.

ESS alone is not sufficient for these targets. It measures calibration on the proposal support, but a molecular generator can miss a conformer and still assign nearly constant weights to the basins it has reached. Evaluation is therefore molecule-specific and always computed from saved particles, so plotting never reruns dynamics. Glycerol and diethanolamine are judged by all torsion marginals, joint rotamer populations, signed-volume populations, energy distributions, and matched-reference reweighting. Alanine dipeptide is additionally judged by its L-only Ramachandran distribution. The FAB million-frame split supplies the fixed alanine reference; matched implicit-solvent references for glycerol and diethanolamine remain part of the pending production workflow.

The reference and generator are compared under exactly the same coordinate convention. Periodic histograms or kernel estimates are constructed only after the raw torsions have been saved, and their smoothing parameters affect the picture rather than the samples. For alanine dipeptide, the familiar  $(\phi, \psi)$  plane is a readable two-dimensional projection of a 60-dimensional measure, not a replacement for full-energy reweighting or the remaining torsions. For the smaller molecules, agreement of one-dimensional marginals is likewise necessary but not sufficient: joint rotamer counts are retained precisely because independently reasonable marginals can conceal different basin weights.

## 6.5 Current status and limitations

The pure-JAX/OpenMM parity tests, mixed-domain flow, MALA, SMC/AIS, checkpointing, and the three frozen bundles pass their bounded smoke tests. Those checks establish implementation consistency and compilation, not sampling quality. The methane and ethane soft-potential runs and the glycerol cutoff scan answer narrower engineering questions: whether the differentiated energy is finite, whether the mixed-domain kernels compile, and whether a proposed softened landscape retains visible torsional modes. They do not establish equilibrium basin weights. The propane sequence, the matched small-molecule references,

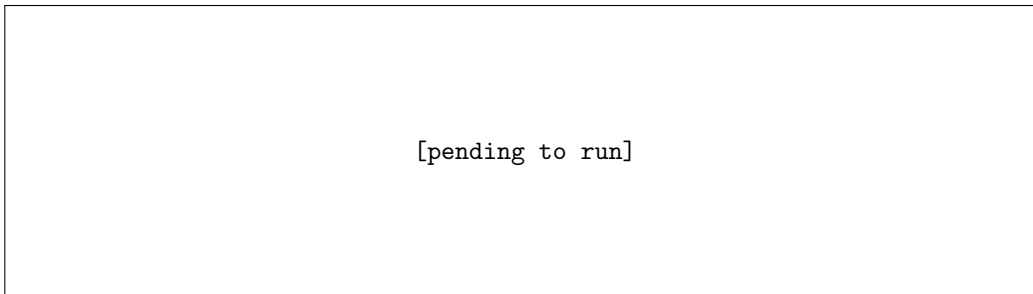


Figure 9: Glycerol implicit-solvent torsion marginals and trained-generator comparison. [pending to run]

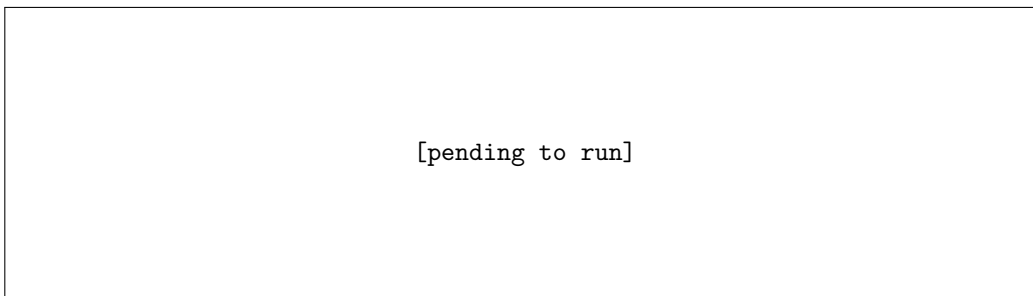


Figure 10: Neutral-diethanolamine implicit-solvent torsion marginals and trained-generator comparison. [pending to run]

and all three production Boltzmann generators remain incomplete. No molecular ESS, ladder factor, or distributional comparison is therefore included in the result claims of this paper revision.

Three limitations define the remaining work. First, the softened starting potential must preserve joint basin weights, not merely one-dimensional torsion marginals. The present glycerol scan shows why: two chains can agree in the plotted marginals and still disagree in their joint rotamer population. Second, alanine dipeptide has a published L-only reference, whereas glycerol and diethanolamine still require independently mixed ensembles under their exact, matched implicit-solvent Hamiltonians. A reference generated under a vacuum model or a different generalized-Born variant would answer a different question. Third, the final comparison must reach the exact potential with unclipped importance weights and must pass the held-out validation-ESS gate at every trained stage. Until those gates pass, the molecular section fixes a reproducible target and protocol but deliberately reports [pending to run] for the learned-sampler panels.

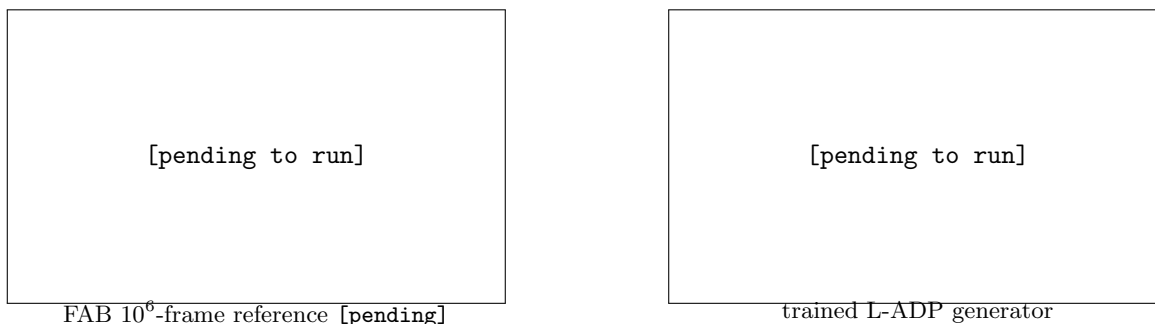


Figure 11: Planned alanine-dipeptide Ramachandran reference and generator comparison on the exact ff96/OBC1 target. [pending to run]

## 7 Conclusions

Forward KL training of a normalizing flow Boltzmann generator usually trains well. But it can miss modes and report a fake effective sample size. Noisy or biased annealed samples limit its accuracy. A smaller leakage failure rides alongside, where the target vanishes. We close these gaps by regularizing the forward KL with a term that reuses the flow’s own log-ratio.

- We propose the log-ratio variation, a family of  $X$  functionals. They augment the forward KL by penalizing the pairwise variation of  $\log(\mu/\nu)$  under a non-negative weight. One forward pass of the flow computes them. The shape term costs little beyond a plain forward KL step; the mixture adds one sampling batch.
- One regularizer takes three roles through its weight.  $X_\mu$  controls the shape of the log-ratio and stabilizes training.  $X_{\hat{\mu}}$  drives mode discovery, drawing the flow toward a wide-coverage measure that quench and temper prepares before training.  $X_{\bar{\nu}}$  suppresses leakage.
- The balanced loss  $KL + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$  combines the three roles in one objective. It folds mode discovery and leakage suppression into a single pairwise integral. It serves as a default and needs little tuning. A Fisher–Rao analysis shows the forward KL still decays along the  $X_\mu$ -regularized flow. Under a bound on the regularization strength, the  $X_\mu$  term can lower the accuracy floor that biased AIS samples leave behind. The forward KL is sensitive to these samples early in training, so the  $X$ -regularized losses help most when the training batch is small. The loss applies across smooth benchmarks, a high-dimensional multi-well, and lattice field theories, where it tends to discover modes and sharpen accuracy.
- We build an adaptive-temperature Boltzmann generator on the loss. It grows its ladder from the SMC effective sample size and evaluates the trained proposal on the complete validation set at every stage. On schedule-matched clock ladders, the balanced loss improves validation ESS on 36 of 37 levels and reduces the cumulative propagation factor at every tested batch size. For molecules, we additionally fix an implicit-solvent Hamiltonian, an exact mixed Euclidean–periodic coordinate measure, and the intended stereochemical support. The molecular production results remain pending and are not included in the empirical claims of this revision.

Several directions invite further study. First, the three fixed molecular benchmarks must be completed against their matched references before comparing with FAB (Midgley et al.); such a comparison would weigh our quench-and-temper coverage source against FAB’s replay buffer. Second, the regularized molecular starting target must preserve joint basin weights while admitting a reliable continuation to the exact potential. Third, larger molecules and explicit solvent require a richer Hamiltonian and reference ensemble, while leaving the mixed-domain sampler unchanged. Finally, the single-run clock comparison should be extended to independent training seeds to quantify its run-to-run variation.

## Acknowledgement

The normalizing flows are built on the JAX packages `jflows` and `jflows_md`, written by the author Xuda Ye. The experiment sources and the two package repositories are available at:

```
https://github.com/xuda-ye-math/X-regularization  
https://github.com/xuda-ye-math/jflows  
https://github.com/xuda-ye-math/jflows\_md
```

The numerical tests run on one NVIDIA GeForce RTX 5090 GPU with 32 GB of VRAM.



## A Fisher–Rao gradient flow with log-ratio variation

Section 2 states the main gradient-flow results used in the paper. This appendix supplies their derivations and proofs on the space of densities. We view  $\nu$  as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities (Ambrosio et al., 2008; Villani, 2009) and ask how fast  $\nu_t$  converges to the target  $\mu$  in  $\text{KL}(\mu\|\nu_t)$ .

In this section we restrict to the loss with only the forward KL and the shape regularizer  $X_\mu$ , the log-ratio variation at weight  $\omega = \mu$ ,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (27)$$

where  $\lambda > 0$  is a fixed constant. The calculations below hold along any positive solution of the Fisher–Rao flow; when  $\lambda \leq \frac{1}{2}$ , the birth term in (33) is non-negative and hence preserves positivity. Throughout this section we abbreviate the importance weight by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is  $z(x) = \log w(x) = \log \mu(x) - \log \nu(x)$ , the field  $z$  of (5). The importance weight  $w$  is distinct from the localization measure  $\omega$  in  $X_\omega$ .

The choice of Riemannian metric matters. In the Wasserstein-2 geometry, induced by optimal transport (Jordan et al., 1998; Otto, 2001), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani / Bakry–Émery machinery (Otto and Villani, 2000; Bakry et al., 2014) that yields exponential convergence for the reverse KL does not apply. We therefore work in the Fisher–Rao (FR) geometry (Amari, 2016; Bauer et al., 2016), in which the forward KL admits an unconditional exponential decay rate and the regularizer  $X_\mu$  contributes a qualitatively new, nonlocal birth–death modulation (Lu et al., 2019): the local update at  $x$  is driven by how the importance weight at  $x$  ranks against the population of weights under  $\mu$ .

Denoting the  $\nu$ -mean by  $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$ , the Fisher–Rao gradient of a functional  $\mathcal{F}[\nu]$  and the associated dissipation identity along the FR gradient flow  $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$  are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left( \frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left( \frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_{\nu_t} \right)^2 \nu_t dx. \quad (28)$$

The mean-subtraction enforces  $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$ , so the FR flow conserves probability mass.

### A.1 Derivation of $X_\mu$ -regularized Fisher–Rao gradient flow

The first variation of the forward KL with respect to  $\nu$  admits the form

$$\frac{\delta \text{KL}(\mu\|\nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (29)$$

so by (28), using  $\langle w \rangle_\nu = \int_{\mathbb{R}^d} \mu dx = 1$ , the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}} [\text{KL}(\mu\|\nu)](x) = \nu(x) - \mu(x). \quad (30)$$

For the  $X_\mu$  functional, the inner log-ratio difference has first variation

$$\frac{\delta (\log w(x) - \log w(y))}{\delta \nu(x)} = \frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)}.$$

By the symmetry of  $X_\mu$  in  $x$  and  $y$ ,

$$\begin{aligned}\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \frac{\delta(\log w(x) - \log w(y))}{\delta\nu(x)} dy \\ &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \left( \frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)} \right) dy \\ &= -2w(x) \int_{\mathbb{R}^d} \mu(y) \operatorname{sgn}(w(x) - w(y)) dy,\end{aligned}$$

where the contribution of  $\delta(x - y)$  vanishes because  $\operatorname{sgn}(w(x) - w(y)) = 0$  at  $y = x$ . Hence

$$\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y \sim \mu} [\operatorname{sgn}(w(x) - w(y))]. \quad (31)$$

We call  $S$  the *rank field* of the importance weight under  $\mu$ :  $S(x) > 0$  indicates that  $w(x)$  is comparatively large within the population of weights, and hence that  $\nu$  pointwise undercovers  $\mu$  relative to that population;  $S(x) < 0$  has the opposite interpretation, and  $S \equiv 0$  at  $\nu = \mu$ . By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x, y \sim \mu} [\operatorname{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of  $X_\mu$  with respect to  $\nu$  is

$$\operatorname{grad}^{\text{FR}}[X_\mu(\mu\|\nu)](x) = -2\mu(x)S(x). \quad (32)$$

Since the FR gradient is linear in the functional, the FR gradient of  $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$  is the direct sum of (30) and (32):

$$\operatorname{grad}^{\text{FR}}[\mathcal{L}](x) = \nu(x) - \mu(x) - 2\lambda\mu(x)S(x).$$

The regularized Fisher–Rao gradient flow  $\partial_t \nu_t = -\operatorname{grad}^{\text{FR}}[\mathcal{L}]$  therefore reads

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda\mu(x)S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[ \operatorname{sgn} \left( \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (33)$$

Equation (33) is a *nonlocal* PDE: its right-hand side at  $x$  depends on  $S_t(x)$ , which in turn requires the entire profile of the importance weight  $\mu/\nu_t$  over the support of  $\mu$ . Mass is conserved along the flow because

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 2\lambda \int_{\mathbb{R}^d} \mu(x) S_t(x) dx = 0,$$

and the flow is stationary at  $\nu_t = \mu$ , where  $\mu(x)/\nu_t(x) \equiv 1$  and hence  $S_t \equiv 0$ . When  $\lambda = 0$ , (33) reduces to the pure Fisher–Rao gradient flow of the forward KL.

## A.2 Convergence analysis of the $X_\mu$ -regularized flow

To analyze the convergence rate of the regularized flow (33), we take the forward KL  $\text{KL}(\mu\|\nu_t)$  as the error metric and differentiate it along the flow:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = - \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx,$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral is settled by the following rank-correlation identity.

**Lemma 1** For any probability density  $\nu$  on  $\mathbb{R}^d$ ,

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|] \geq 0,$$

with equality if and only if  $w$  is  $\mu$ -almost-everywhere constant.

*Proof.* Since  $\mu^2/\nu = \mu w$ , the definition of  $S$  gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_{\mu} [w(x) S(x)] = \mathbb{E}_{x,y \sim \mu} [w(x) \operatorname{sgn}(w(x) - w(y))].$$

Symmetrizing in  $x \leftrightarrow y$  gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [(w(x) - w(y)) \operatorname{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|].$$

□

Applying Lemma 1 with  $\nu = \nu_t$  and  $S = S_t$ , the dissipation rewrites as

$$\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) = -\chi^2(\mu \| \nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[ \left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (34)$$

The elementary inequality  $w \log w \leq w^2 - w$  applied to  $w = \mu/\nu_t$  gives  $\chi^2(\mu \| \nu_t) \geq \operatorname{KL}(\mu \| \nu_t)$ , hence  $\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) \leq -\operatorname{KL}(\mu \| \nu_t)$ , and Grönwall's lemma yields the exponential decay

$$\operatorname{KL}(\mu \| \nu_t) \leq e^{-t} \operatorname{KL}(\mu \| \nu_0), \quad t \geq 0, \quad (35)$$

with no structural assumption on  $\mu$ : neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

The uniform convergence rate (35) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR flow is a continuous-time idealization on the space of densities and is agnostic to the sampling cost. In practice, the difficulty of training has been partially transported to the AIS step: evaluating the forward KL and its gradient requires accurate samples of  $\mu$ , and producing such samples in high dimensions is itself a hard problem. This is the reason that AIS is needed, and that the  $X_{\hat{\mu}}$  regularizer must compensate for modes that AIS alone fails to reach. The exponential rate (35) quantifies how fast the flow *can* converge given accurate target samples; it does not eliminate the burden of producing them.

A naive comparison of dissipation rates between KL-only and  $\operatorname{KL} + \lambda X_{\mu}$  would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of  $X_{\mu}$  is structural rather than rate-based: while the pure KL FR gradient is local in  $x$ , the rank field  $S_t$  is global, so the regularized flow (33) couples the update at  $x$  to the entire distribution of importance weights under  $\mu$ . This nonlocal coupling amplifies the birth rate on undercovered regions of  $\mu$  and suppresses it on overcovered ones, a global comparison that pure forward KL, which knows only whether  $\mu(x)/\nu_t(x)$  exceeds or falls below 1 pointwise, cannot produce.

Two further cautions concern the discrete realization of the flow. First, the time-reparametrization argument cuts in both directions once the optimizer enters. Training follows Adam steps, not the FR flow: the per-coordinate second-moment normalization absorbs a global rescaling of the loss, so a larger gradient buys no speedup by itself, while a regularizer that reshapes the relative magnitudes of the gradient components does change the optimizer's effective time scale. Part of any observed difference in training speed between KL and  $\operatorname{KL} + \lambda X_{\mu}$  may therefore be optimizer scaling rather than information geometry, and we claim no universal transient speedup on this basis. Second, the dissipation (34) is a population statement, while each gradient step sees an estimate from a finite, generally biased target surrogate. Small batches are the primary regime for which the regularizer was designed. The staged clock experiment tests this finite-batch regime with a schedule-matched comparison: its quantitative claims use the saved trained-flow, full-validation ESS at each level, not a composed endpoint estimate or a training-minibatch diagnostic.

**Remark (Monotone decay of  $X_\mu$ )** The regularizer  $X_\mu$  also decreases monotonically along the regularized flow: pairing the variation  $\delta X_\mu / \delta \nu = -2wS$  (31) against the flow (33) gives

$$\frac{d}{dt} X_\mu(\mu \| \nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0,$$

with strict inequality unless  $\mu/\nu_t$  is  $\mu$ -almost-everywhere constant, the first term being non-positive by Lemma 1 and vanishing only in that case. Thus  $X_\mu$  is itself driven to zero in the long-time limit.

### A.3 Asymptotic accuracy under biased AIS samples

The convergence analysis of Section A.2 assumes that the outer expectation in  $\text{KL}(\mu \| \nu) + \lambda X_\mu(\mu \| \nu)$  is taken under the exact target  $\mu$ . In practice, the algorithm only has access to an approximation  $\tilde{\mu}$  of  $\mu$  supplied by AIS, and the realized loss reads

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (36)$$

where the pointwise ratio  $w(x) = \mu(x)/\nu(x)$  is still evaluated against the exact target, since it is computed per sample from the target potential  $U$  and the flow's own density. Repeating the derivation of Section A.1 with  $\mu$  replaced by  $\tilde{\mu}$  in the outer integrands gives the Fisher–Rao gradient

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda \tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}} [\text{sgn}(w(x) - w(y))].$$

A stationary point  $\nu^*$  of (36) therefore satisfies the self-consistent equation

$$\nu^*(x) = \tilde{\mu}(x) \left[ 1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \text{sgn} \left( \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right) dy \right]. \quad (37)$$

At  $\lambda = 0$  the equation collapses to  $\nu^* = \tilde{\mu}$ : pure forward KL training inherits the bias of the target surrogate  $\tilde{\mu}$  verbatim, and the residual error is exactly  $\text{KL}(\mu \| \tilde{\mu})$ . For  $\lambda > 0$ , however, the rank-based correction  $1 + 2\lambda \tilde{S}^*$ , where  $\tilde{S}^*$  denotes  $\tilde{S}$  evaluated at  $\nu = \nu^*$ , inflates  $\nu^*$  above  $\tilde{\mu}$  where the importance weight  $\mu/\nu^*$  is high, that is, where  $\nu^*$  still undercovers  $\mu$ , and deflates it where the weight is low, providing a global feedback that pulls  $\nu^*$  from  $\tilde{\mu}$  back toward  $\mu$ . Two facts prepare the statement. For  $\lambda < \frac{1}{2}$  the bracket in (37) is strictly positive, because  $\tilde{S}^* \in [-1, 1]$ . Moreover, the antisymmetry of the sign in  $x, y$  implies

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \tilde{S}^*(x) dx = \mathbb{E}_{x, y \sim \tilde{\mu}} \left[ \text{sgn} \left( \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right) \right] = 0,$$

hence every stationary point  $\nu^* = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)$  is a probability density.

**Theorem 2** Assume that  $\mu$  is absolutely continuous with respect to  $\tilde{\mu}$  and that  $\lambda \in [0, \frac{1}{2})$ . Then every stationary density  $\nu^*$  of the loss  $\tilde{\mathcal{L}}$  (36) satisfies

$$\text{KL}(\mu \| \nu^*) \leq \text{KL}(\mu \| \tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (38)$$

*Proof.* Substituting the stationary identity  $\nu^* = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)$  into the difference of forward KLs and using  $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)(\mu/\nu^*)$ ,

$$\text{KL}(\mu \| \tilde{\mu}) - \text{KL}(\mu \| \nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda \tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} u(x) \log u(x) dx,$$

where  $u(x) := 1 + 2\lambda \tilde{S}^*(x) \geq 1 - 2\lambda > 0$ , since  $\tilde{S}^* \in [-1, 1]$  (a  $\tilde{\mu}$ -average of signs) and  $\lambda < \frac{1}{2}$ . The elementary inequality  $u \log u \geq u - 1$  for  $u > 0$  then yields

$$\text{KL}(\mu \| \tilde{\mu}) - \text{KL}(\mu \| \nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

Applying the rank-correlation argument of Lemma 1 verbatim with  $\mu$  replaced by  $\tilde{\mu}$  in the outer measure,

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (38).  $\square$

The right-hand side of (38) (Theorem 2) contrasts the two roles of  $X_\mu$  training. At  $\lambda = 0$  the bound is tight at  $\nu^* = \tilde{\mu}$  and the residual KL is fixed at  $\text{KL}(\mu \parallel \tilde{\mu})$ ; the surrogate bias is a hard floor. Switching on the regularizer with  $\lambda < \frac{1}{2}$  lowers this floor (strictly, unless  $\mu/\nu^*$  is constant on the support of  $\tilde{\mu}$ ) by an amount controlled by the discrepancy of the importance weights  $\mu/\nu^*$  across the support of  $\tilde{\mu}$ : the larger this discrepancy, the more the rank-based feedback pulls  $\nu^*$  toward  $\mu$ . The  $X_\mu$  regularizer is therefore not merely a numerical stabilizer for the forward KL flow: within the theorem’s assumptions, it lowers the asymptotic accuracy floor of the trained density when the target surrogate is biased.

The threshold  $\lambda < \frac{1}{2}$  is a sufficient condition enforcing strict positivity of  $\nu^*$  uniformly in  $x$ . The experiments use  $\lambda = 1$  and show robust performance, but they do not extend this accuracy-floor theorem beyond its stated range. Within that range, the result shows that  $X_\mu$  alone can reduce the bias floor inherited from an inaccurate target surrogate.

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